

A marked tensor obstruction to four-dimensional disk embedding

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Abstract

We disprove the unrestricted four-dimensional disk-embedding conjecture. We construct immersed disks in a compact oriented smooth four-manifold with framed algebraic dual spheres satisfying the usual equivariant intersection and reduced self-intersection conditions, but with no pairwise disjoint locally flat replacements that preserve the boundary maps and induced normal framings. The obstruction holds even when the replacement disks' relative homotopy classes are not prescribed.

1 Introduction

The disk-embedding problem asks when immersed disks in a four-manifold can be replaced by disjoint locally flat embedded disks with the same boundary data. Its role in four-dimensional topological surgery is to turn algebraic cancellation data into embedded disks. In dimension four, the auxiliary disks used to carry out cancellations can themselves intersect the original surfaces and one another. Dual spheres provide additional control. An algebraic dual records one intersection with its designated disk and zero with the others, counted with signs and fundamental-group labels; a geometric dual realizes these counts by meeting its designated disk once transversely and missing the other disks.

Casson's construction replaces the finite task of removing intersections by an infinite tower of immersed disks. Freedman's recognition theorem identifies the resulting Casson handle, relative to its attaching region, with the standard open topological two-handle. Together these results established the simply connected foundation of disk embedding [8, Introduction and Theorem 1.1]. Freedman and Quinn developed the capped-grope version of this method and organized the disk theorem and its surgery and cobordism applications [9, Theorem 5.1A and Chapter 11]. The group restriction enters through the double-point loops of the caps: their null-homotopies supply further stages of the construction. The class of *good groups* is defined by the availability of the required π_1 -null disk operations [15, Definition 3.2]. Freedman and Teichner proved that groups of subexponential growth are good [10, Theorem 0.1]. Krushkal and Quinn supplied another proof [13, Theorem, p. 408]. Their paper also contains Freedman and Teichner's correction of the earlier height-raising argument [13, Appendix, pp. 424–430].

The modern formulation of Powell, Ray, and Teichner supplies framed geometric dual spheres as well as embedded disks, and preserves the homotopy classes of the input dual spheres [15, Theorem A]. Their modification supplies the geometric-dual step missing from the earlier argument; the disk conclusion without these output spheres was already proved [15, Remark 1.3]. For generically immersed input disks, their theorem preserves the induced boundary normal framings without prescribing relative homotopy classes for the replacement disks. The *unrestricted disk-embedding*

conjecture considered here is the extension of that theorem obtained by removing only the good-group hypothesis. It would in particular give framed disk replacements for compact oriented smooth input manifolds, even after the output-dual requirement is discarded. We resolve this geometric conjecture negatively by obstructing that disk conclusion.

We use the following intersection conventions in stating the result. After choosing basepoints and whiskers, λ denotes the equivariant intersection pairing with values in $\mathbb{Z}[\pi_1 M]$. For an oriented four-manifold M , the reduced self-intersection invariant $\tilde{\mu}$ takes values in the additive quotient

$$\mathbb{Z}[\pi_1 M] / \langle h - h^{-1} \ (h \in \pi_1 M), \ \mathbb{Z} \cdot 1 \rangle.$$

A framed immersed sphere has a trivialized normal bundle. A boundary normal framing is the framing induced by the normal bundle of the specified immersed disk.

Theorem 1.1. *There exist a compact connected oriented smooth four-manifold M with boundary, an integer $k \geq 1$, generically immersed proper disks*

$$f_1, \dots, f_k : (D^2, S^1) \longrightarrow (M, \partial M)$$

with disjoint embedded boundary circles, and generically immersed framed spheres $g_1, \dots, g_k : S^2 \longrightarrow M$, such that

$$\lambda(f_i, g_j) = \delta_{ij}, \quad \lambda(g_i, g_j) = 0 \quad \text{for all } i, j, \quad \tilde{\mu}(g_i) = 0 \quad \text{for all } i,$$

but there are no pairwise disjoint locally flat embedded disks in M with the same boundary maps and the boundary normal framings induced by the f_i .

The diagonal intersection equations are part of the statement. The conclusion of Theorem 1.1 requires no homotopy constraint on a hypothetical replacement disk. The markings used in the proof specify based curves and their paths in the fixed construction; they impose no further condition on hypothetical replacement disks. The result concerns geometric disk replacement. The relation to independently formulated surgery assertions is discussed after the proof in Section 7.

1.1 Idea of the proof

We first construct a fixed immersed disk problem in the exterior of a finite collection of capped groves. A capped grove here is a surface tree with disks attached at its terminal curves; its boundary records an iterated commutator in the terminal curves. The construction installs finitely many such commutator relations, with independently chosen based conjugations, in a smooth plumbing model. It also supplies the algebraic dual spheres required in Theorem 1.1. These data are fixed before any hypothetical embedded replacements are chosen.

Suppose this particular disk problem has replacements with the required boundary maps and framings. They allow the plumbing model to be cut into two vertex pieces A, B , joined by compact three-manifolds $Y_1, \dots, Y_{2m}$ with torus boundary, for an integer $m \geq 5$ chosen in the construction. The cuts are smooth and miss the locally flat disks. The installed boundary words are now relations in the vertex groups, and restriction in cohomology gives complementary linear data on the two sides. Proposition 2.2 records this marked geometry; Lemma 2.4 verifies the equivariant disk-and-sphere input.

For each edge Y_i , we study based flat SL_2 -connections with one torus holonomy fixed. Basing records holonomy as a matrix, with no quotient by simultaneous conjugation. A *potential* is a function in finitely many formal coordinates whose gradient equations describe these flat connections.

The *central potential* fixes this torus holonomy to the identity. Proposition 4.5 constructs it from the Chern–Simons functional, including a first-order variation of the fixed boundary holonomy. Theorem 5.1 then replaces the central potential by an algebraic function through a formal coordinate change. The comparison preserves based holonomy on the full critical scheme: its coordinate ring is the power-series ring modulo the actual gradient ideal, with its nilpotent elements retained. The geometric reduction selects m edges for the first-order boundary variations; we call these edges active.

The two vertex pieces determine formal graphs in the product of the edge coordinate spaces. Their central tangent spaces are complementary, and suitable sums of the edge potentials vanish identically on the graphs. The boundary relations imply more than vanishing at common critical points: products associated to every three distinct active edges belong to the restricted gradient ideals. On one side these products are $t_i t_j t_k$, where each $t_i^2 = 0$ and mixed products remain in the parameter ring. On the other they are products of three algebraic holonomy functions. Proposition 6.1 establishes the exact vanishing and ideal memberships. The graph parametrizations may mix all edge coordinates, while each fixed potential depends only on its own edge block.

Theorem 3.1 rules out precisely these algebraic data when at least five edges are active. Artin approximation and finite specialization first reduce them to large positive characteristic. Truncating each internal coordinate by its characteristic power makes the associated Koszul complexes finite. The two transverse graphs determine classes with nonzero pairing. The target triple relations, together with ordinary exterior differentiation, decompose one class into tensors admitting three independent first-order parameter lifts. The source triple relations force each resulting pairing to vanish, giving the contradiction.

Three features are useful beyond the final contradiction. First, the geometry imposes a finite family of independently conjugated laws before the cuts are chosen. Second, the algebraicity argument compares full relative closed shifted forms and preserves the marked holonomies; an algebraic model of the reduced representation locus would be insufficient. Third, the tensor argument uses actual gradient ideals over nonreduced parameter rings and makes no isolated-critical-point or regular-sequence assumption.

1.2 The geometric and algebraic antecedents

Capped-grope contraction and transverse grope constructions provide the local geometric operations [13, §§1.2 and 1.4]. Their use to relate surface intersections and commutator words also appears in the work of Freedman and Krushkal. Using the repeated-variable 2-Engel relation, they constructed homotopy solutions to the A–B slice problem and new universal surgery models [6, Theorems 1 and 2]; their later work established universality for half- π_1 -null surgery models, with a capped grope on one side and a π_1 -null sphere on the other [7, Theorem 1 and Section 3]. These results concern particular surgery or slicing formulations, with their own embedding and complement conditions. They explain the value of commutator calculus in this setting without supplying the marked geometric reduction used here.

Our relations use three distinct group inputs and become trivial when any one is the identity. Their logarithmic expansion begins with the trilinear bracket $[Z_1, [Z_2, Z_3]]$, where Z_a is the logarithm of the a -th input. The deletion property is the Brunnian property defined in Section 2. We install a finite family of such words with independently chosen conjugations, before knowing the cut pieces. The later argument uses those independent choices to recover every triple coefficient. It neither imposes a universal Engel relation nor requires the vertex groups to be nilpotent. The simultaneous placement, framing, and algebraic-dual calculations needed for this finite family are given in Section 2.

The deformation-theoretic part has a different origin. Goldman and Millson describe Artin-algebra families of flat connections by differential graded Lie algebras and identify their based holonomies with representation deformations [12, Corollary 6.4 and Proposition 6.6]. Their quadraticity results use additional formality hypotheses in the compact Kähler setting. The edge manifolds here are arbitrary compact three-manifolds, so higher deformation terms remain. Chern and Simons constructed the classical transgression form whose exterior derivative is the corresponding curvature characteristic form [4, Section 3, especially Proposition 3.2]. Section 4 uses this form to construct the potential, computes its boundary adjustment, and proves equality of its full curvature and gradient ideals.

The algebraic model comes from shifted symplectic geometry. The Betti transgression theorem of Pantev, Toën, Vaquié, and Vezzosi produces shifted symplectic forms on local-system stacks; their Lagrangian intersection theorem lowers the shift by one [14, Theorems 2.5 and 2.9]. Calaque’s relative orientation construction supplies the Lagrangian restriction map associated to an oriented manifold with boundary [3, Theorem 2.9 and §3.1]. After a punctured filling of the edge, these results put us in the setting of Brav, Bussi, and Joyce’s algebraic Darboux theorem in shift -1 , which supplies an algebraic critical chart [2, Example 5.15 and Theorem 5.18(i)].

To use that chart, we must identify its full relative closed form with the form underlying the particular Chern–Simons potential already constructed. Proposition 5.2, proved in Appendix A, carries out this comparison on derived coefficients and source cochain models. Its inputs include Getzler’s simplicial Maurer–Cartan comparison for nilpotent Lie algebras [11, Theorem 5.4 and Corollary 5.11] and the simplicial de Rham contraction of Dupont [5]. Section 5 then turns the closed-form comparison into a coordinate change fixing the critical scheme and its based holonomies. Only after this step, and the polynomial replacement of the holonomy logarithms, do we apply ordinary Artin approximation [1, Theorem 1.10]. The approximation is applied to the final graph equations with their fixed, separately algebraic edge functions.

1.3 Conventions and organization

We assume familiarity with basic four-manifold topology, cohomology with local coefficients, and formal deformation theory. The algebraicity argument additionally uses derived mapping stacks and shifted symplectic forms; its required comparison is stated in the body and proved in full in Appendix A.

All manifolds and maps in the initial model are smooth, with corners rounded when necessary. Hypothetical output disks are only locally flat. The subsequent cuts are smooth hypersurfaces in the original smooth manifold and miss those disks. Thus the vertex and edge calculations use ordinary smooth forms and Stokes’ theorem.

Unless a field is specified, cohomology has characteristic-zero coefficients. We use $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ and the pairing $\kappa(X, Y) = \text{tr}(XY)$. Connections and coordinate changes are formal power series, coefficient by coefficient. A statement about an ideal is a statement in the complete coordinate ring, including its nilpotents. In Sections 4–6, d denotes the differential on source forms and δ differentiation in formal coordinates. In the tensor argument of Section 3, d is the coordinate de Rham differential and δ its induced operator on Koszul homology. We specify total signs when using both differentials.

Section 2 proves the geometric reduction. Section 3 states and proves the finite obstruction before we construct its inputs, making the target algebraic interface explicit. Sections 4 and 5 construct the edge germs and their algebraic models. Section 6 constructs the vertex graphs and checks every hypothesis of the tensor obstruction. Section 7 completes the proof of Theorem 1.1.

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2 The geometric reduction

We use the disk assertion only to compress a finite collection of surfaces. The following restricted version of its disk conclusion is therefore sufficient. All manifolds in the constructions before that application are smooth and oriented; corners are rounded when necessary. The output will be a fixed disk problem with algebraic duals, together with a description of the cut manifolds that its hypothetical embedded solutions would produce. The marked boundary words and their based paths are chosen as part of the initial disk problem.

Assumption 2.1 (Universal framed disk replacement). *Let M be a compact connected oriented smooth four-manifold with boundary. Suppose that a finite collection of proper generic immersions*

$$f_i: (D^2, S^1) \longrightarrow (M, \partial M), \quad 1 \leq i \leq k,$$

has pairwise disjoint embedded boundary circles, and admits generically immersed framed spheres $g_1, \dots, g_k$ such that

$$\lambda(f_i, g_j) = \delta_{ij}, \quad \lambda(g_i, g_j) = 0, \quad \tilde{\mu}(g_i) = 0 \quad \text{for all } i, j.$$

Here intersections are counted in $\mathbb{Z}[\pi_1 M]$, with fixed whiskers, the diagonal equations are included, and

$$\tilde{\mu}(g_i) \in \mathbb{Z}[\pi_1 M] / \langle h - h^{-1}, \mathbb{Z} \cdot 1 \rangle$$

uses the additive quotient. Then the f_i can be replaced by pairwise disjoint locally flat embedded disks with the same boundary maps and the same induced normal framings on those boundary circles.

No relative homotopy class for the replacement disks is prescribed in Assumption 2.1. Nor does it require output dual spheres. The unrestricted version of [15, Theorem A] would imply this assumption, including its framing clause. We will contradict the assumption itself. The capped-surface contractions used below are the usual one-sided contractions; see [13, §1.2] and [15, Lemma 3.4 and its preceding discussion]. The relation between grope trees and commutator words is also used in [6, 7]. We give the particular finite construction needed here.

2.1 A finite list of relations

Our commutator convention is $[a, b] = aba^{-1}b^{-1}$. A word R in three free generators X_1, X_2, X_3 will be called a *three-slot grope word* if it is the boundary word of the rooted surface tree consisting of a punctured torus with a single tip on its first branch and a second punctured torus on its other branch. Paths to its three tips are part of its marking. Changing those paths inside the body can conjugate a tip by words in the three based tips, and can conjugate the inner commutator by such a word. Overall conjugation of the boundary word is irrelevant to its being a relation. In particular,

$$R|_{X_a=1} = 1 \quad (a = 1, 2, 3), \quad R \equiv [X_1, [X_2, X_3]] \pmod{\Gamma_4 F(X_1, X_2, X_3)}. \quad (2.1)$$

Here $\Gamma_r F$ denotes the lower central series of the free group F . The first property is the *Brunnian property in the three slots*: setting any one input equal to the identity trivializes the word. The orientations of the two surfaces can be fixed to give the displayed sign. Indeed a tip conjugation changes its image modulo Γ_2 by zero, and hence changes this weight-three commutator only modulo Γ_4 . The first assertion follows directly by deleting a branch of the rooted tree. Equivalently, in the completed free associative algebra over $\mathbb{Q}$,

$$\log R(e^{Z_1}, e^{Z_2}, e^{Z_3}) = [Z_1, [Z_2, Z_3]] + \text{terms of total degree at least four}; \quad (2.2)$$

each term contains a letter from every slot. These two properties, rather than a claim that an entire vertex group is nilpotent, will be used.

Fix $m \geq 5$ and $p \geq 9$. Each plumbing block will have $2m$ solid-torus boundary regions, called *ports*, through which it joins other blocks. Removing these regions leaves $2m$ boundary tori. Write $x_1, \dots, x_{2m}$ for their meridians and $l_1, \dots, l_p$ for the extra one-handle generators, and put

$$\mathcal{C} = \{1, l_1, \dots, l_p\}.$$

Before making any cut, install one marked grope body for every ordered triple of distinct port indices, every independent choice $c_1, c_2, c_3 \in \mathcal{C}$, and every independent choice $\delta_1, \delta_2, \delta_3 \in \{1, -1\}$. Its three based tips represent

$$X_a = c_a x_{r_a}^{\delta_a} c_a^{-1}, \quad a = 1, 2, 3. \quad (2.3)$$

There are at most

$$8(2m)(2m-1)(2m-2)(p+1)^3$$

such bodies. Any additional fixed finite list of words in the extra letters could be included in $\mathcal{C}$ by the same construction. In particular, neither the list of bodies nor their markings depends on the slopes selected by the cuts below.

Proposition 2.2 (Geometric reduction). *For the integers m, p and the finite list just specified, there is a compact connected oriented smooth four-manifold X , constructed without Assumption 2.1, with the following properties.*

It has a regular cover with deck group F_m , whose block graph is the Cayley tree of F_m . Each block is a four-dimensional one-handlebody on the p extra letters. Its exposed boundary piece is

$$K_v \cong (\#^p(S^1 \times S^2)) \setminus \text{int } \nu(L_{2m}),$$

where L_{2m} is a $2m$ -component unlink in a ball. The port meridians and the extra letters freely generate $\pi_1 K_v$. At every join, the meridian on one side is the longitude on the other. There is a finite family of framed grope bodies with boundary in these pieces, equivariant in the cover, realizing the list (2.3).

If Assumption 2.1 holds, there are equivariant cuts, with compact quotient, having connected vertex pieces V_v and connected edge three-manifolds Y_e . Each Y_e has as its entire boundary the prescribed join torus. Every corresponding three-slot grope word is trivial in $\pi_1 V_v$.

In the index-two parity quotient, these cuts give two connected vertex pieces A, B and $2m$ connected oriented edge pieces $Y_1, \dots, Y_{2m}$. Their ambient manifold X_2 is homotopy equivalent to the graph with two vertices and $2m$ edges, with p additional circles at each vertex. For any characteristic-zero field k , difference of restrictions gives

$$H^2(A; k) \oplus H^2(B; k) \xrightarrow{\cong} \bigoplus_{i=1}^{2m} H^2(Y_i; k). \quad (2.4)$$

In degree one it is surjective, and restricts to an isomorphism on the kernel of evaluation on the p extra loops at each vertex.

At each boundary torus choose the two original coordinate loops α_i, β_i so that α_i has nonzero image in $H_1(Y_i; k)$ and β_i has zero image. Such a choice is possible: one of the original coordinate directions, not just an arbitrary slope, is killed. Call the endpoint where β_i is a longitude the plus endpoint; there α_i is the port meridian. There are exactly m edges with plus endpoint A . After numbering these as $1, \dots, m$, the finite relation list includes, at A , all the required three-slot relations on any three distinct α_i , and, at B , those on the corresponding β_i , with independent conjugations by $\mathcal{C}$.

The proof constructs a particular exterior E and a finite disk family in it before making any cut. A simultaneous framed replacement of that family alone implies all the cut conclusions of the proposition; the universal Assumption 2.1 is used only to provide that replacement. Lemma 2.4 specifies the fixed family and verifies its algebraic duals. This local implication will give the fixed counterexample in Theorem 1.1.

2.2 The plumbing blocks

Start with a standard four-dimensional one-handlebody with $2m + p$ one-handles. Attach a cancelling two-handle to each of the first $2m$ one-handles, using the product framing and disjoint standard attaching regions. Before plumbing, the resulting block B is a one-handlebody on the remaining p handles. The belt circles of the cancelling two-handles are a standard unlink in a ball in ∂B . One can see each component separately from the cancelling pair: the complement of its attaching solid torus in $S^1 \times S^2$ is a solid torus, which becomes the complement of the belt unknot after cancellation. The framed attaching pushoff is its belt meridian. The cancellation regions for different pairs are disjoint.

Pair the $2m$ two-handles. In each handle choose a small product patch $D_{\text{base}}^2 \times D_{\text{fibre}}^2$, and plumb the patches of its pair by interchanging the two factors. This is mutual plumbing; there are no self-plumbings. To specify the local manifold structure, the two branches at a plumbing can be represented by

$$(D_R^2 \times D_r^2) \cup (D_r^2 \times D_R^2) \subset \mathbb{R}^2 \times \mathbb{R}^2, \quad 0 < r < R.$$

Their overlap is the product ball $D_r^2 \times D_r^2$. Rounding its corners gives the usual smooth plumbing chart. Distinct patches are disjoint. The factor interchange preserves the orientation in dimension four, and the product framings fix the signs of the resulting transverse core intersections.

Equivalently, take one copy B_v for each vertex of the Cayley tree of $F_m = \langle a_1, \dots, a_m \rangle$, and identify the designated patches of B_v and B_{va_j} by this chart. Neighbouring blocks intersect in one product ball, and there are no other or triple intersections. Left translation acts freely. The quotient is the finite plumbing X , and the union is its regular cover $\tilde{X}$. Since all attaching intersections are contractible cofibrations, the homotopy type of the union is obtained by joining the block homotopy types along the tree. Thus

$$\tilde{X} \simeq \text{the Cayley tree with } p \text{ circles attached at each vertex.} \quad (2.5)$$

This is not an assertion that $\tilde{X}$ is the universal cover: the extra circles remain.

At a product patch the removed part of the unplumbed block boundary is a solid-torus neighbourhood of the belt circle. Its boundary is $S_{\text{base}}^1 \times S_{\text{fibre}}^1$. Consequently the exposed boundary is exactly the piece K_v in Proposition 2.2, with a torus collar at each join. The factor interchange exchanges the two peripheral slopes. An unlink longitude bounds a disk in K_v , whereas its meridian is one of its free generators. Adding the p extra one-handles gives the remaining free generators l_s .

Fix boundary whiskers for these generators. In the cover, fix the boundary map to the tree which is constant on each K_v core and crosses the corresponding tree edge on every torus collar. The middle of each collar maps to the midpoint of that edge. The construction is equivariant. It descends both to the one-vertex graph and to the parity quotient.

2.3 Marked grope bodies and their framings

The plumbing fixes the port meridians and peripheral slopes. We next place the finite list of surface bodies with their prescribed based tip curves, and match their framings to the handle cores. These choices must be made before the exterior disk problem is defined.

Lemma 2.3 (The marked body model). *The two-stage grope body used in (2.1) has a three-dimensional handlebody thickening which, with its three marked tip annuli, is a boundary join of three longitudinal-annulus collars. Finitely many copies can be placed disjointly in the boundary of the initial one-handlebody, with prescribed parallel tip tracks to the cancelling-handle attaching regions and with the external conjugations (2.3). Their surface framings can be matched to the handle-core framings.*

Proof. We first identify the thickening together with its tip annuli, then place the finite marked family, and finally match the surface framings to the cap framings while keeping the terminal attaching circles fixed.

1. Primitive tip annuli. Thicken a punctured torus S to $S \times [-1, 1]$. Put its two selected basis curves on opposite faces of this product. There are disjoint proper cutting arcs in S , each dual to one basis curve and disjoint from the other. For an explicit model, puncture the square torus near $(1/2, 1/2)$, take the basis circles at $y = 1/4$ and $x = 1/4$, and take the cutting arcs at $x = 1/2$ and $y = 1/2$, with their ends on the puncture. The only intersection of the latter two arcs was removed by the puncture. Their products with the interval are disjoint meridian disks, each meeting just its own marked tip once.

For the lower stage, view the primitive b -annulus as the longitudinal annulus of a solid torus. That solid torus is an annular collar, up to rounding, and the rest of the lower handlebody is the spare a -handle. Glue the annular boundary of the upper punctured-torus product to this b -annulus. Gluing the collar only thickens that annulus on the upper handlebody. Extend the two upper meridian disks across the collar. Their traces occupy finitely many strips there. Place the feet and return arc of the spare a -handle in a complementary interval. Its meridian disk is then disjoint from both extended upper disks. We have obtained three disjoint disk duals for the three marked tips.

This is a statement about the marked annuli, not just about a free basis. Cut the handlebody along the three dual disks. Each marked annulus becomes a rectangle on the boundary of the resulting ball, joining only the two faces of its own cutting disk. The rectangles are disjoint. A neighbourhood of each rectangle together with its two disk faces is a disk in the boundary sphere, disjoint from the other two such disks. Smooth their corners. A smooth isotopy of the boundary sphere carries these three labelled disk neighbourhoods to standard ones; within each neighbourhood, choose it also to standardize the two disk faces and their connecting rectangle. This last choice is an isotopy of a disk containing two subdisks joined by a band. Extend the boundary isotopy over the ball by smooth isotopy extension. Reattaching the handles therefore identifies the *marked* smooth handlebody with a boundary join of the three longitudinal-annulus collars.

2. Finite placement with based paths. Reserve, on the attaching regions in the initial boundary, disjoint annuli of product parallels for every tip in the finite list. Join their thin collars by thin tubes along the desired based graph paths, avoiding the reserved approaching strips. Finitely many arcs in a three-manifold can be made disjoint, apart from the prescribed common ends of a graph; parallel copies separate repeated portions. Regular neighbourhoods therefore give the required disjoint marked handlebodies. A path winding along an extra handle changes a based tip by the prescribed external conjugation while leaving its free attaching curve a parallel of the same cancelling-handle attaching loop. Paths internal to the marked body can add only the core-word changes already allowed in (2.1).

3. Matching the cap framings. The placement fixes the external paths and the terminal product circles at the two-handles. We now vary the winding of each surface ribbon and its return collar to match the cap's normal framing, without moving those terminal circles or paths. These choices precede the definition of the exterior disk problem.

In an oriented three-dimensional body crossed with an interval, frame a surface normal bundle by its oriented normal in the body and the interval vector. At a tip, a cap collar departing in the first normal direction has normal frame consisting of the in-surface direction transverse to the tip and the interval vector. Transport this ordered pair with the collar to the attaching region. The discrepancy with the two-handle core's product frame is an integer. It can be changed on the relevant solid-torus handle by

$$(\theta, z) \mapsto (\theta, e^{in\theta} z), \quad n \in \mathbb{Z}.$$

The core stays fixed and the marked longitude changes by n meridians. The three disk duals make these choices independent. To realize the winding change with the same terminal product circle, choose the return collar after making the twist. Use coordinates $S_\theta^1 \times D_z^2 \times [0, 1]_u$ around a separated tip and its return collar. A twisted surface ribbon has local form $(\theta, te^{in\theta}, 0)$. Its outgoing normal in the three-dimensional body is $ie^{in\theta}$. Choose an embedded plane arc $(a(s), b(s))$, $0 \leq s \leq 1$, with $a(0) = a(1) = b(0) = 0$, $b(1) = 1$, $a'(0) > 0$, $b'(0) = 0$, and $a = 0$, $b' > 0$ near $s = 1$. We may require $b' > 0$ for $0 < s \leq 1$. Then

$$(\theta, s) \mapsto (\theta, a(s)ie^{in\theta}, b(s))$$

is an embedded cap collar. It leaves the ribbon in the required outgoing direction and ends at the same product circle $z = 0, u = 1$, independently of n . In its normal quotient bundle a frame is given by

$$v_n = e^{in\theta}, \quad w_n = a'(s)\partial_u - b'(s)ie^{in\theta}.$$

The two vectors are independent modulo the tangent plane; orthogonal projection gives a normal frame if desired. At the initial circle this is the surface transverse vector together with the collar vector. At the terminal circle it is, up to a constant change of frame, $(e^{in\theta}, ie^{in\theta})$, whose winding relative to the product core frame is n . Thus the framing discrepancy changes by an arbitrary integer while the terminal cap circle stays fixed. If a handle foot is to remain fixed, replace $e^{in\theta}$ by a degree- n circle map constant near that foot. The disjoint tip neighbourhoods allow all these choices simultaneously. The construction uses the separated tip representatives on the opposite faces of the surface products, not disjointness of the unpushed symplectic basis curves.

Choose the integers to remove the framing discrepancies. This transports the body and its marked annuli together; it does not add a relative Euler term to a previously fixed cap. Meridians are null in the handlebody, so the core words and their leading commutator terms are unchanged. The twists are fixed near the connector feet, so the external paths realizing the c_a are unchanged. Any internal change of path is a word $W(X_1, X_2, X_3)$ in the three abstract tips of the marked handlebody. Under the external marking $X_a = c_a x_{r_a}^{\delta_a} c_a^{-1}$, this word specializes to $W(1, 1, 1) = 1$ when the three port meridians are trivial, even if the extra-letter holonomies are not trivial. Thus such path changes do not alter the degree-three term in (2.2).

Finally put the bodies at an inner level of a short boundary collar, with their initial boundary and three tip tracks returning to the outer level in disjoint collars of the four marked curves. The initial boundary avoids the attaching regions. The two directions at a symplectic pair depart on opposite normal sides. Continue the tip tracks into separate product parallels of the two-handle cores. These caps miss every body away from their own attachments. Their only possible intersections are the core intersections in the specified plumbing charts. $\square$

In particular the finite family is constructed without knowing the future cut pieces. Each cap is individually embedded: a core has no self-plumbing. Caps of different labels may intersect, and no disjointness of their spanning surfaces or of all their interiors is asserted.

2.4 The disk problem in the exterior

The marked bodies have now been placed with framed caps. Removing only their bottom surfaces leaves an exterior in which one cap per body is the input disk. The remaining upper stage will supply its algebraic dual sphere; the unused upper cap will identify the group-ring labels needed for cancellation.

Denote the disjoint bottom punctured tori by $S_1, \dots, S_k$. Choose their open product tubes, including their proper boundary collars, and put

$$E = X \setminus \bigcup_{j=1}^k \text{int } \nu(S_j).$$

The tubes are chosen after the finite placements of Lemma 2.3. They can be sufficiently thin that they avoid all the upper caps and meet the upper bodies only in their short attaching collars. The exterior is connected: a path can be moved off finitely many proper codimension-two surfaces and then off their sufficiently small tubes. It is a compact oriented smooth four-manifold after rounding.

Lemma 2.4 (Algebraic duals in the exterior). *There are proper generically immersed disks f_j in E , with disjoint boundary circles and with the bottom-surgery framings, and framed spheres g_j in E , such that*

$$\lambda(f_i, g_j) = \delta_{ij}, \quad \lambda(g_i, g_j) = 0, \quad \tilde{\mu}(g_j) = 0.$$

All equalities hold over $\mathbb{Z}[\pi_1 E]$.

Proof. Write $D_{x,j}$ for the single cap on the a -side of S_j , and use as f_j its proper truncation on $\partial\nu(S_j)$. Its product framing is the framing required for compression of the bottom.

Let U_j be the upper punctured torus. Write $D_{y,j}$ for its chosen surgery cap and $D_{z,j}$ for the other cap. Compress U_j along $D_{y,j}$. Cutting U_j along its y -curve gives a pair of pants; fill its two new boundary components by opposite parallel copies of $D_{y,j}$. The result is a disk

$$C_j = U_j^{\text{cut}} \cup D_{y,j}^+ \cup (-D_{y,j}^-) \quad (2.6)$$

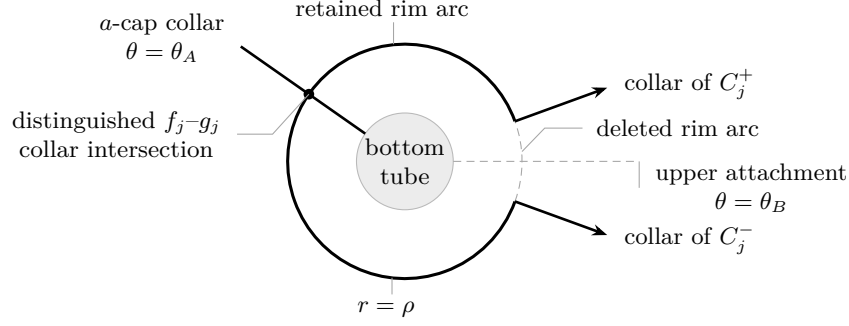
with boundary a parallel of b_j . Its collar is on the outgoing side of the upper attachment. The unused $D_{z,j}$ is not part of C_j : it will instead give a null-homotopy in E identifying the group-ring labels of the two surgery-cap sheets.

Take the normal-circle torus over a parallel of b_j near $\partial\nu(S_j)$, and compress it with two parallels of C_j . Here is a collar model fixing the relevant separations. Use coordinates (s, t, r, θ) , where s is along b_j , t is transverse to it on S_j , and (r, θ) are polar coordinates normal to S_j . The torus is $t = 0$, $r = \rho$. The a -cap collar and the upper attachment depart at distinct angles θ_A, θ_B . Remove the thin compression band at θ_B , keep the annulus containing θ_A , and place the interiors of the two compression disks outside radius ρ . Thus

$$g_j = R_j^{\text{cut}} \cup C_j^+ \cup (-C_j^-). \quad (2.7)$$

The retained annulus meets the a_j -collar in one transverse point. Orient and base it so that this point contributes 1. All other body pieces, approach tracks and collars are in the separated neighbourhoods of Lemma 2.3. In particular no collar point occurs for f_i, g_j with $i \neq j$. Figure 1 shows the normal-disk cross-section of this compression. Restoring the s -coordinate turns the retained arc into the annulus and the deleted arc into the compression band.

We count all remaining intersections. They occur in cap-cap plumbing charts, because the parent bodies and the collar pieces just described are disjoint away from their prescribed attachments.



Normal-disk cross-section; the coordinates s and t are suppressed.

Figure 1: The collar of the compressed rim sphere near the crossing of a_j and b_j on the bottom surface. Restoring the coordinate s along b_j turns the solid rim arc into the retained annulus, the dashed rim arc near θ_B into the removed compression band, and the two outgoing segments into collars of the compression disks. The marked point locates the distinguished collar intersection of f_j with g_j ; the suppressed transverse surface coordinate is t . Only local collar directions are shown. The interiors of $C_j^\pm$, the other caps, and their possible mutual intersections are not depicted.

Each occurrence of a $D_{y,j}$ sheet in (2.7) has two indices: $\varepsilon \in \{+1, -1\}$ specifies the copy of C_j , and $\eta \in \{+1, -1\}$ specifies the cap sheet inside that copy. Its orientation coefficient is $\varepsilon\eta$; the two indices are shown in Figure 2. At a plumbing chart with coordinates $(u, v) \in D^2 \times D^2$, these sheets are small parallel coordinate disks $v = v_{\varepsilon,\eta}$; the opposite core sheets are $u = u_{\varepsilon',\eta'}$. For any fixed opposing sheet, their local intersection signs differ by the factor η .

Fix ε . The two paths to the η -sheets have the same stem from the rim torus to that copy of the upper body. Their difference is, up to this common conjugation, the upper basis loop z_j . That loop bounds $D_{z,j}$, together with its tip-annulus collar, in E . Indeed those upper caps were disjoint from all the bottom tubes; intersections with other caps do not remove them from E . An infinitesimal normal turn around the upper body also bounds a normal fibre disk in E , since that body was not removed. It follows that the two intersection labels in $\pi_1 E$ agree. The paired contributions are therefore

$$\varepsilon\eta h + \varepsilon(-\eta) h = 0 \quad \text{in } \mathbb{Z}[\pi_1 E].$$

The potential winding around the *bottom* normal circle distinguishes the outer ε -copies. We never compare those copies in this cancellation: ε is held fixed throughout.

This accounts for every cap contribution to $\lambda(f_i, g_j)$. For g_i, g_j , fix a sheet of g_i and the outer index of g_j , and cancel the inner index of g_j by the same argument. It also applies when all parallel sheets at a plumbing point are counted simultaneously. Hence the off-diagonal pairings of the g_j vanish.

The diagonal and framing admit a direct geometric check. The disk C_j uses just the one core type $D_{y,j}$, which has no self-plumbing. That cap and its parallels are embedded and miss the parent body off their attachments. Thus C_j , and then the sphere in (2.7), can each be individually embedded. Mutual intersections with other g_i do not affect this assertion. The normal frame of the rim torus is (∂_t, ∂_r) . Along the outgoing radial compression collar the cap normal frame is $(\partial_t, \partial_\theta)$. Rounding the surgery corner rotates the second vector from ∂_r to ∂_θ , while the first remains fixed. The product framing matched in Lemma 2.3 extends this ordered pair. The earlier compression (2.6) is the same framed operation, so its outgoing b_j -annulus has exactly the required inherited framing. Consequently g_j has trivial normal bundle. Its normal push-off is disjoint from it, proving

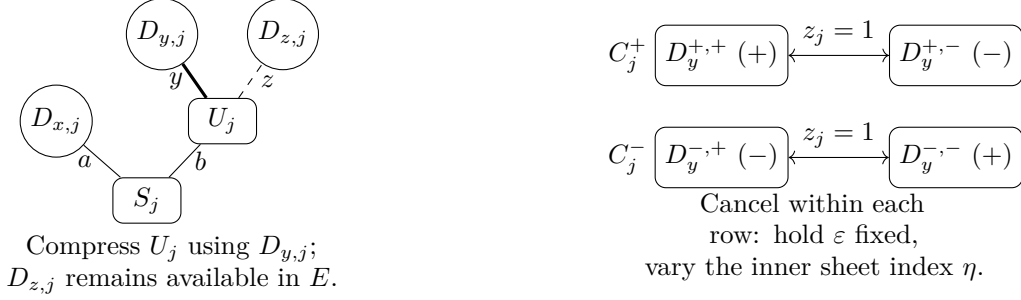


Figure 2: Combinatorial schematics, not projections of the four-dimensional embedding. Left: the two-stage body and its one-sided upper contraction. Right: the four occurrences of its surgery cap in the compressed rim sphere; parentheses record their orientation coefficients $\varepsilon\eta$. The right panel suppresses the cap's subscript j . The labels $z_j = 1$ are equalities in $\pi_1(E)$, proved using the upper dual cap in E . No comparison between the two rows, which could involve a bottom normal-circle meridian, is used.

the diagonal $\lambda(g_j, g_j) = 0$; its embedded representative also gives $\tilde{\mu}(g_j) = 0$.

Combining this with the distinguished collar point proves all the stated equalities. Small perturbations make the finite collection generic while preserving the calculation and the framings. $\square$

Assumption 2.1, applied to E and Lemma 2.4, now gives pairwise disjoint locally flat embedded replacement caps with the specified boundary framings. Compress each bottom S_j with two parallels of its replacement cap, using the returning annuli in the removed tube. The result is a collection of pairwise disjoint properly embedded disks in X , bounded by the initial grope boundaries. The framing is what makes this compression locally flat with the required product collar. No relative homotopy of the replacement cap to f_j is used. The output dual-sphere part of a disk embedding theorem is not used either.

2.5 Equivariant cuts with connected pieces

The preceding compression uses the assumed replacement disks to make the initial grope boundaries bound disjoint disks. We now keep those disks in their prescribed vertex regions while cutting the ambient plumbing into smooth pieces. Connectedness of these pieces will be needed for the restriction maps in the final part of this section.

Lemma 2.5 (Connected tree cuts). *Suppose the initial grope boundaries bound the disjoint disks just constructed. Then the boundary map to the plumbing tree extends equivariantly to a map whose cuts can be chosen to have the connected vertex and edge pieces asserted in Proposition 2.2. Every lifted disk lies in its prescribed vertex piece.*

Proof. Lift the disks to $\tilde{X}$. Each is labelled by the vertex containing its boundary. Distinct translates and distinct members have disjoint images, because the disks were embedded and disjoint in X . Prescribe the map to be constant at the label vertex on a neighbourhood of every such disk, and retain the fixed boundary map. These prescriptions agree on their overlaps.

An equivariant extension is equivalently a section of the associated bundle over X with fibre the tree. The prescribed section extends first to a neighbourhood of the prescribed closed set, since the tree is an absolute neighbourhood retract, and then over the remaining cells, since the fibre is contractible. One may take a finite triangulation of the smooth ambient manifold for the latter extension after shrinking the neighbourhood. The section formulation ensures equivariance. This

step does not require extending to the noncontractible quotient graph by contractibility of that graph.

On the quotient, approximate the map smoothly over the open middle intervals of its finitely many graph edges, keeping it fixed on the specified boundary collars and away from the disk neighbourhoods. Choose regular values there. Their inverse images are compact smooth proper hypersurfaces, disjoint from all disks. After rescaling an edge interval we call its chosen value the midpoint. The lift of the level corresponding to a tree edge e has exactly one boundary component: the specified torus T_e . More precisely, the unique component meeting the boundary contains that torus as its entire boundary; every other component of that level is closed.

Consider the component graph $\mathcal{T}'$ of this cut in $\tilde{X}$: its vertices are connected complementary pieces, and its edges are connected cut hypersurfaces. It is connected. The collapse to $\mathcal{T}'$ is surjective on fundamental groups: a finite graph loop is realised by choosing crossing points and joining successive points by paths in the connected pieces. By (2.5), the fundamental group of $\tilde{X}$ is generated by the extra loops in its boundary pieces, with whiskers. Each such loop maps to a vertex of $\mathcal{T}'$. Thus $\pi_1 \mathcal{T}' = 0$, and $\mathcal{T}'$ is a tree.

The boundary pieces and their torus collars determine a principal copy $\mathcal{T}$ of the original tree inside $\mathcal{T}'$. To check injectivity directly, remove the chosen midpoints from the target tree. A connected component of their inverse-image complement maps into just one resulting vertex-star component. It cannot contain both K_v and K_w for $v \neq w$. Similarly distinct principal edge components have distinct midpoint labels. The boundary collars give exactly the required incidences. Hence the principal copy is indeed an embedded subtree.

Every remaining branch of $\mathcal{T}'$ has a unique nearest vertex in $\mathcal{T}$. A deck transformation stabilising a vertex or edge of such a branch fixes that nearest principal vertex and is therefore the identity. Thus all these stabilisers are trivial. The finite compact quotient cut has only finitely many components, so $\mathcal{T}'$ has finite quotient. Trivial stabilisers imply finite valence. Moreover distance to $\mathcal{T}$ is invariant under translation and is bounded on the finite set of component orbits. A finite-valence tree of bounded depth rooted at one principal vertex is finite.

Ignore the closed cuts in each such finite hanging branch, merging it with its principal vertex piece. This operation is equivariant, preserves compact quotient and the principal hypersurfaces, and gives connected vertex pieces V_v and a single connected edge piece Y_e with $\partial Y_e = T_e$. All prescribed disks remain in their own pieces: they lay in the component containing their boundary before the merging. The Y_e and V_v are smooth manifolds with corners; the topological disks themselves were never required to become smooth. $\square$

Each initial boundary is therefore null-homotopic in its vertex piece. Its actual boundary word has the three-slot properties (2.1) with inputs (2.3). This proves precisely the relation statement in Proposition 2.2.

2.6 The parity quotient and its peripheral homology

Send every plumbing generator of F_m to $1 \in \mathbb{Z}/2$. The quotient of $\tilde{X}$ by the kernel has two principal vertex pieces A, B and two lifts of each of the m plumbing edges. Thus there are $2m$ edges. The free translation interchanging parity gives a deck involution interchanging A and B , and pairing these edge lifts. Equation (2.5), or the same contractible-overlap argument in this quotient, gives

$$X_2 \simeq \Gamma \text{ with } p \text{ extra circles at each vertex,} \quad |V\Gamma| = 2, \quad |E\Gamma| = 2m. \quad (2.8)$$

In particular $H^q(X_2; k) = 0$ for $q \geq 2$, and $\dim_k H^1(X_2; k) = 2m - 1 + 2p$. The extra loops at the two vertices can be assigned independently in the free ambient fundamental group. All port meridians are trivial there, since they bound the corresponding handle-core disks in X_2 .

Lemma 2.6 (Restriction maps and coordinate slopes). *The cohomology and slope conclusions of Proposition 2.2 hold. For each edge piece one also has*

$$H^1(Y_i, \partial Y_i; k) \cong \ker(H^1(Y_i; k) \longrightarrow H^1(\partial Y_i; k)), \quad b_2(Y_i) = b_1(Y_i) - 1. \quad (2.9)$$

Proof. Insert small edge collars so that Mayer–Vietoris applies to the two vertex pieces with intersection the disjoint union of the Y_i . Its degree-one portion is

$$\begin{aligned} H^0(A; k) \oplus H^0(B; k) &\longrightarrow \bigoplus_i H^0(Y_i; k) \longrightarrow H^1(X_2; k) \\ &\longrightarrow H^1(A; k) \oplus H^1(B; k) \xrightarrow{R} \bigoplus_i H^1(Y_i; k) \longrightarrow 0. \end{aligned}$$

Connectedness makes the first cokernel have dimension $2m - 1$; these are the graph classes of Γ . They restrict to zero on the vertex pieces. The remaining kernel of R has dimension $2p$. Evaluation on the p extra loops at each vertex identifies this kernel with k^{2p} : their ambient classes in (2.8) give surjectivity, and an ambient class with zero extra evaluations is a graph class and restricts to zero. Consequently R restricts to an isomorphism on the kernel of the extra evaluations. The next part of the same sequence, using $H^2(X_2; k) = H^3(X_2; k) = 0$, gives (2.4).

Let Y be one of the connected oriented three-manifolds with single torus boundary T . Poincaré–Lefschetz duality and $\chi(Y) = \chi(T)/2 = 0$ give $b_2(Y) = b_1(Y) - 1$. The exact sequence of the pair, using connectedness of Y, T , gives the isomorphism in (2.9). Equivalently the image of $H_1(T; k)$ in $H_1(Y; k)$ has dimension one, the half-lives/half-dies statement.

Write u, v for the original two coordinate directions on T . One is a longitude in the boundary piece on one side and the other is a longitude on the other side. Suppose both had nonzero image in the one-dimensional image in $H_1(Y; k)$. They would then be nonzero scalar multiples of each other. A cohomology class restricted from the first side annihilates its longitude and hence both directions; the same is true of a class restricted from the second side. Surjectivity of R would imply that every class of $H^1(Y; k)$ annihilates both directions, contradicting the one-dimensional image. Thus one original coordinate direction has zero image and the other has nonzero image. Denote them by β, α , respectively. In cohomology the boundary restriction image is the line spanned by the coordinate form dual to α .

The deck involution preserves the transported slope choice and swaps the two vertices. For the two lifts of each plumbing edge it therefore interchanges which one has its plus endpoint at A . Exactly one lift in each pair is active, giving exactly m active edges. At A their α ’s are distinct port meridians, and at B their β ’s are distinct port meridians. Their three-slot relations, with every independent choice of the stipulated conjugators, were all included before any slope was selected. $\square$

Proof of Proposition 2.2. The plumbing construction supplies X , its boundary pieces, and its regular cover. Lemma 2.3 supplies the finite marked family. Lemma 2.4 checks every input hypothesis of Assumption 2.1 in the exterior of the bottoms. The replacement disks and framed bottom compressions then give the disks required by Lemma 2.5, and hence the relations in connected vertex pieces. Finally Lemma 2.6 gives the claimed parity-quotient data. $\square$

3 A finite tensor obstruction

This section proves an algebraic obstruction to the formal data that will arise from the two vertex pieces. The formal parametrizations may involve every edge block; each potential depends only on its own block. All parameter rings below retain every mixed square-free parameter monomial.

For a characteristic-zero field K , write $K\langle h \rangle$ for the ring of algebraic power series at the origin, identified with the henselization of $K[h]_{(h)}$ inside $K[[h]]$. An algebraic germ can therefore be represented by a regular function on a pointed étale neighborhood. Gradients in this section always differentiate internal coordinates, keeping the external parameters fixed.

Theorem 3.1 (Finite tensor obstruction). *Let $h = (h_1, \dots, h_n)$ be a finite list of coordinate blocks over a characteristic-zero field K , of total dimension N . Let $I \subset \{1, \dots, n\}$ have cardinality $m \geq 5$, and let*

$$F_i \in K\langle h_i \rangle \quad (1 \leq i \leq n), \quad b_i \in K\langle h_i \rangle \quad (i \in I).$$

Put

$$T = K[t_i : i \in I]/(t_i^2 : i \in I), \quad S_0(h) = \sum_{i=1}^n F_i(h_i), \quad S_t(h) = S_0(h) + \sum_{i \in I} t_i b_i(h_i).$$

Suppose there are formal parametrizations

$$H_L(x, t) \in (x, t) T[[x]]^N, \quad H_R(y) \in (y) K[[y]]^N,$$

where x has a coordinates and y has $N - a$ coordinates, with the following properties.

(i) The central tangent maps

$$\partial_x H_L(0, 0) : K^a \longrightarrow K^N, \quad \partial_y H_R(0) : K^{N-a} \longrightarrow K^N$$

are injective and have complementary images.

(ii) The identities

$$S_t(H_L(x, t)) = 0, \quad S_0(H_R(y)) = 0$$

hold in $T[[x]]$ and $K[[y]]$, respectively.

(iii) For every three distinct $i, j, k \in I$,

$$t_i t_j t_k \in ((\partial_{h_\nu} S_t)(H_L(x, t)) : 1 \leq \nu \leq N) \subset T[[x]], \quad (3.1)$$

$$(b_i b_j b_k)(H_R(y)) \in ((\partial_{h_\nu} S_0)(H_R(y)) : 1 \leq \nu \leq N) \subset K[[y]]. \quad (3.2)$$

Then these data do not exist.

Condition (i) makes the parametrizations smooth formal graph embeddings, including over the full ring T . Nilpotent constant displacements of H_L are allowed. The ideals in (3.1)–(3.2) are the actual gradient ideals; no passage to their radicals is made. There is no flatness assumption on their quotients.

The later construction supplies the hypotheses as follows; all triples in the table have distinct active indices.

Input	Source of the data	Later verification
Separate algebraic $F_i(h_i)$ and $b_i(h_i)$	Individual marked edge models and polynomial perturbations	Theorem 5.1 and Section 6.7
Complementary central tangents	Restriction maps after fixing the prescribed evaluations	Lemma 6.2
Exact vanishing $S_t _L = 0$ and $S_0 _R = 0$	Stokes and section corrections preserving the residual quotient	Lemmas 6.3, 6.4, and 6.7
Every source triple $t_i t_j t_k$ in the restricted gradient ideal on L	Boundary laws at A	Lemma 6.6
Every target triple $b_i b_j b_k$ in the restricted gradient ideal on R	Boundary laws at B and polynomial holonomy weights	Lemma 6.6 and Section 6.7

The threshold $m \geq 5$ comes from two counts. The target triple identities put the right graph's Koszul class in a sum of tensor patterns with at most two active factors outside the multiplication kernels. A compatible projection for the induced exterior differential then gives pure tensors of classes with at least $m - 2 \geq 3$ active factors both closed for that differential and killed by b_i . These give three independent parameter lifts, and the source triple membership forces their pairing to vanish. The closure statements concern Koszul classes; the chosen form representatives need not be closed for ordinary d .

3.1 Algebraic approximation and finite specialization

Lemma 3.2. *If the data in Theorem 3.1 existed, data satisfying the same conditions would exist over a field of positive characteristic p , for arbitrarily large p . The edge functions would still be separated in their original blocks.*

Proof. First put the parametrizations into graph form. Choose complementary linear projections adapted to their central tangent spaces. For L , the chosen projection of H_L has invertible x -Jacobian modulo (x, t) . It therefore has a formal inverse over T , with the parameters fixed. Its constant displacement is in (t) , so this is a legitimate substitution in the complete local ring $T[[x]]$. For R the same assertion is the ordinary formal inverse function theorem. After these reparametrizations each graph has a fixed linear projection equal to its independent variables.

We describe explicitly the approximation problem for L . For $A \subset I$, let $t_A = \prod_{i \in A} t_i$, including $t_\emptyset = 1$. Expand its graph functions and all the coefficients witnessing (3.1) in this finite basis:

$$H_L(x, t) = \sum_{A \subset I} t_A H_A(x), \quad c(x, t) = \sum_{A \subset I} t_A c_A(x).$$

All unknown coefficient functions have the same list x of independent variables.

The fixed algebraic functions F_i, b_i admit finite pointed étale presentations. In such a presentation, differentiation in h_ν extends uniquely from the polynomial coordinate ring: implicit differentiation expresses the derivative of every represented function using the inverse étale Jacobian. Thus the functions and their ambient first derivatives can be handled in one finite presentation. Introduce auxiliary unknowns for its coordinates and its inverse units, and expand them too in the basis t_A . Equating coefficients in the following identities gives a finite polynomial system:

- the étale presentation equations and the inverse-unit equations;
- the fixed graph-projection equation;
- $S_t(H_L) = 0$;
- each equation expressing $t_i t_j t_k$ as a linear combination of the restricted ambient gradient entries.

No derivative of an unknown graph function occurs. The only derivatives are the fixed ambient derivatives of the given potentials, represented as just described. The existing formal graph and membership witnesses give a formal solution of this polynomial system.

Artin approximation over the henselization $K[x]_{(x)}^h$ [1, Theorem 1.10] supplies an algebraic solution preserving the desired finite jets. Preserve the centers, the branch centers of the étale presentations, and the central first graph jet. Do the same independently for R , encoding (3.2) in its own independent variables y . These two systems share only fixed functions. Their compatibility condition is the open condition of complementary central tangents, which the preserved jets retain. There is consequently no nested approximation problem. If one independent-variable list is empty, the corresponding system is already a finite system over K .

We now have finitely many algebraic functions and finitely many exact identities in henselizations. Each function is represented on a finite pointed étale neighborhood. An identity in a henselization holds at some finite stage of the filtered system of these neighborhoods. Enlarging the stages finitely many times represents all the functions, compositions, membership identities, branch centers, and inverse units together. Their finite presentations descend to a finitely generated integral $\mathbb{Z}$ -subalgebra $A \subset K$, after adjoining the required inverse units. In particular we retain invertibility of the étale Jacobians and of the transverse tangent determinant. The implicit differentiation formulas descend as well, so the specialized gradient entries remain the derivatives of the specialized potentials. The original presentations for the edge functions involve only their own blocks h_i , and retain that property after descent.

For completeness, this finite localization still has closed points in arbitrarily large positive characteristics. The nonzero finite-type $\mathbb{Q}$ -algebra $A \otimes_{\mathbb{Z}} \mathbb{Q}$ has a maximal ideal with residue field a number field E . The images of the finitely many generators of A , including all inverted elements, belong to $\mathcal{O}_E[1/d]$ for some positive integer d . Reduction at any prime of $\mathcal{O}_E$ over a rational prime not dividing d gives a homomorphism from A to a finite field. All the declared units remain units. The reduced pointed étale presentations give formal germs over that residue field, satisfying the same identities and the same tangent condition. $\square$

Remark 3.3. The lemma also applies when $K = \mathbb{C}((\sigma))$. Only finitely many elements of K occur as coefficients of the finite presentations, and each is treated as a single coefficient. Specialization is performed on A , not coefficientwise on Laurent series or on the whole field K . In particular, no boundedness of the negative σ -orders of the coefficients of an Artin solution is required. The algebraicity of the fixed separated potentials is essential: the argument does not assert that ordinary Artin approximation can algebraize arbitrary unknown formal potentials while preserving separate variable restrictions.

3.2 Truncated forms and transverse graph classes

The approximation step preserves all graph identities and actual ideal memberships. We now turn their positive-characteristic specializations into identities in finite-dimensional complexes, where transverse graphs can be tested by a nonzero pairing.

Fix a specialization supplied by Lemma 3.2 in characteristic $p > 2$, and denote its field by $\mathbb{k}$. Continue to use the same letters for the specialized germs. Write $z_1, \dots, z_N$ for all internal coordinates and set

$$A_p = \mathbb{k}[z_1, \dots, z_N]/(z_1^p, \dots, z_N^p), \quad \Omega_p = A_p \otimes_{\mathbb{k}} \bigwedge_{\mathbb{k}} \langle dz_1, \dots, dz_N \rangle.$$

This is the ordinary algebra of differential forms of A_p . Its differential is well defined because $d(z_i^p) = 0$. With external parameters, use

$$T_p = \mathbb{k}[t_i : i \in I]/(t_i^2 : i \in I), \quad \Omega_{p,T} = \Omega_p \otimes_{\mathbb{k}} T_p,$$

and differentiate only in the internal coordinates. All these spaces are finite over the indicated coefficient rings.

Lemma 3.4. *Every formal change of internal coordinates over T_p that is centered modulo (t) preserves the ideal $(z_1^p, \dots, z_N^p)$. Hence it induces an automorphism of the truncated algebra and of its relative differential forms.*

Proof. If $c \in (z, t)T_p[[z]]$, Frobenius shows that c^p belongs to $(z_1^p, \dots, z_N^p)$: every nonconstant internal monomial acquires an exponent divisible by p , and every positive parameter monomial has zero p -th power. Apply this to each new coordinate. The inverse coordinate change has the same property, giving equality of the ideals and the asserted automorphisms. $\square$

Define a $\mathbb{k}$ -linear functional on forms, zero outside exterior degree N , by

$$\int f(z) dz_1 \wedge \dots \wedge dz_N = [z_1^{p-1} \dots z_N^{p-1}]f.$$

Extend it T_p -linearly in the presence of parameters. It vanishes on exact forms: a derivative contributing exponent $p-1$ in its differentiated variable would have to come from exponent p , whose derivative is zero. Consequently, for homogeneous forms,

$$\int d\alpha \wedge \beta = -(-1)^{|\alpha|} \int \alpha \wedge d\beta. \quad (3.3)$$

Let $c_1, \dots, c_r$ be graph equations for a smooth formal graph over the parameter ring. Their joint center is zero modulo (t) . Its graph form is

$$U_c = (c_1^{p-1} dc_1) \wedge \dots \wedge (c_r^{p-1} dc_r) \in \Omega_{p,T}^r. \quad (3.4)$$

This expression involves only a finite jet after truncation.

Lemma 3.5. *The form U_c is d-closed and is annihilated by the graph ideal. Every internal one-form whose restriction to the graph is zero wedges to zero with U_c . For two central graphs with complementary tangents and graph forms U_0, V ,*

$$\int U_0 \wedge V \neq 0. \quad (3.5)$$

Proof. Each factor $c_j^{p-1} dc_j$ is closed, and $c_j^p = 0$ in the truncated ring by Lemma 3.4. This proves the first two assertions. In graph coordinates, the kernel of restriction of internal one-forms is the sum of graph-ideal multiples of one-forms and the span of the dc_j . Both summands wedge to zero with U_c . This description remains valid after truncation because the graph coordinate change preserves the truncation ideal.

For the pairing, concatenate the central graph equations for the two graphs, and let M be their linear coefficient matrix in the original coordinates. Complementarity says that M is invertible. The coefficient part of $U_0 \wedge V$ has degree at least $N(p-1)$; all terms of higher degree vanish in A_p . Thus only the linear terms of the graph equations and the constant terms of their differentials contribute. A linear coordinate change M acts on the polynomial socle $\mathbb{k}z_1^{p-1} \cdots z_N^{p-1}$ by $\det(M)^{p-1}$, and on the exterior top form by $\det(M)$. To check the socle assertion one may use diagonal matrices, permutations, and elementary transvections: for a transvection every nonconstant binomial term has an exponent at least p and vanishes. These matrices generate the general linear group. The pairing is therefore $\det(M)^p$, up to the choice of order of the graph equations, and is nonzero. $\square$

Let U_t be the graph form for L , and let V be the graph form for R . Define the degree-one operators

$$D_t = dS_t \wedge (-), \quad D = dS_0 \wedge (-).$$

Both square to zero. The graph identities and Lemma 3.5 give

$$D_t U_t = 0, \quad DV = 0, \quad dU_t = 0, \quad dV = 0. \quad (3.6)$$

In particular $DU_0 = 0$ and the nonzero pairing (3.5) is defined.

Lemma 3.6. *For every three distinct active indices i, j, k , there are actual forms B_{ijk} and C_{ijk} with*

$$t_i t_j t_k U_t = D_t B_{ijk}, \quad b_i b_j b_k V = DC_{ijk}. \quad (3.7)$$

The first equality is over the full parameter algebra T_p .

Proof. Lift the graph membership coefficients to the ambient ring. The difference between the asserted scalar and the resulting ambient gradient combination belongs to the graph ideal, so it annihilates the graph form. If ι_ν is contraction with the coordinate vector $\partial/\partial z_\nu$, then

$$\iota_\nu D_t + D_t \iota_\nu = (\partial_{z_\nu} S_t) \text{ id}.$$

For a D_t -cycle U_t , multiplication by $\sum_\nu a_\nu \partial_{z_\nu} S_t$ is consequently

$$\left(\sum_\nu a_\nu \partial_{z_\nu} S_t \right) U_t = D_t \left(\sum_\nu a_\nu \iota_\nu U_t \right).$$

The operator D_t does not differentiate the coefficients a_ν . This proves the first equality, and the same argument with D, V proves the second. $\square$

We will use two elementary facts about this pairing. If θ is a one-form and $D_\theta = \theta \wedge (-)$, then, for homogeneous α ,

$$(D_\theta \alpha) \wedge \beta = (-1)^{|\alpha|} \alpha \wedge D_\theta \beta. \quad (3.8)$$

Thus a D_θ -boundary wedges to zero with a D_θ -cycle, even before integration. Also $dD + Dd = 0$, since $d^2 S_0 = 0$. Ordinary differentiation therefore induces a differential on the homology of D ; Equation (3.3) continues to test its boundaries against actual closed representatives.

3.3 Two homologies and a compatible projection

We have graph cycles with a nonzero pairing and actual boundary identities for every triple. The separate edge blocks now let us analyze the target graph class one factor at a time. We first take Koszul homology, then use the differential induced on it by ordinary exterior differentiation.

Let Ω_i be the truncated form algebra in the block h_i , and write

$$D_i = dF_i \wedge (-), \quad H_i = H(\Omega_i, D_i).$$

The tensor products of graded spaces and operators in what follows use the usual Koszul signs. The block separation gives

$$(\Omega_p, D) = \bigotimes_{i=1}^n (\Omega_i, D_i), \quad H(\Omega_p, D) = \bigotimes_{i=1}^n H_i.$$

The second equality is the finite-dimensional Künneth isomorphism over the field $\mathbb{k}$. Let δ_i denote the differential induced on H_i by ordinary d , and let δ be their tensor differential. For $i \in I$, multiplication by b_i induces a degree-zero endomorphism B_i of H_i , since it commutes with D_i . Set

$$K_i = \ker B_i.$$

There is no assertion that K_i is a δ_i -subcomplex.

Lemma 3.7 (Kernel support). *Let $v \in \bigotimes_{i=1}^n H_i$ be killed by $B_i B_j B_k$ for every three distinct active indices. Then v belongs to the sum of tensor subspaces having all but at most two active factors in the corresponding K_i .*

Proof. Choose a graded complement J_i to K_i in each active factor. The restriction of B_i to J_i is injective. For a fixed triple, the kernel of its tensor operator is the sum of subspaces having a K_i factor in at least one of the three specified positions: the induced tensor of injections on the quotient by those kernels is injective. Expand the whole tensor product using $H_i = K_i \oplus J_i$. Intersecting the kernel conditions for all triples leaves exactly the summands with at most two active J_i positions. $\square$

Lemma 3.8 (Projection preserving kernel representatives). *Let (H, δ) be a finite-dimensional graded complex over a field, and let $K \subset H$ be any graded subspace. There is a chain projection $P : H \rightarrow H$, homotopic to the identity, whose image is a space of representatives for $H(H, \delta)$ and which satisfies*

$$P(K) \subset K \cap \ker \delta.$$

Proof. Write $Z = \ker \delta$ and $B = \text{im } \delta$. Choose a complement A_K to $K \cap B$ in $K \cap Z$, and extend A_K to a complement A to B in Z . Choose a complement W_K to $K \cap Z$ in K . It is disjoint from Z , so extend it to a complement W to Z in H . All these choices are made degree by degree. Then

$$H = B \oplus A \oplus W.$$

Let P be projection onto A . It is a chain map, and $P(K) \subset A_K \subset K \cap Z$. The map $\delta : W \rightarrow B$ is an isomorphism with the appropriate shift of degrees. Define a degree-minus-one map s to be its inverse on B , and zero on $A \oplus W$. Then

$$\text{id} - P = \delta s + s \delta.$$

$\square$

Apply Lemma 3.8 to each active factor (H_i, δ_i, K_i) , and choose an ordinary homology projection in each inactive factor. Denote these maps by P_i , and put $\Pi = \bigotimes_i P_i$. This tensor projection is chain-homotopic to the identity. Indeed, the telescoping identity for $\text{id} - \bigotimes_i P_i$, together with the individual homotopies and the graded tensor convention, supplies a homotopy for the total differential δ .

Let $v = [V] \in H(\Omega_p, D)$. Equations (3.7) imply the hypotheses of Lemma 3.7, and $\delta v = 0$ by (3.6). Thus Πv is a sum of homogeneous pure tensors

$$v_1 \otimes \cdots \otimes v_n \quad (3.9)$$

such that every v_i is a δ_i -cycle and at least $m - 2$ active factors lie in $K_i \cap \ker \delta_i$. This conclusion follows by applying Π to the tensor subspaces in Lemma 3.7; the projection of a kernel factor is still a kernel factor. Since $m \geq 5$, every nonzero pure tensor in this sum has at least three such active factors.

Choose D_i -cycle representatives for all the factors in (3.9), and let V' be the resulting sum of actual forms. The chain homotopy for Π shows that $[V'] - [V] = \delta[Z]$ for some class in $H(\Omega_p, D)$. Choose a D -cycle representative Z . There is then an actual form E such that

$$V' - V = dZ + DE. \quad (3.10)$$

Pairing with U_0 , equations (3.3), (3.6), and (3.8) give

$$\int U_0 \wedge V' = \int U_0 \wedge V. \quad (3.11)$$

The representatives composing V' need not be closed for ordinary d . Their classes are δ_i -cycles, which is precisely what was used to obtain (3.10).

3.4 Lifts over a three-parameter cube

The replacement class has the same pairing with the source graph as the original target class. Each of its tensor summands has at least three active factors lying in both the multiplication kernel and the kernel of the induced differential. Those three factors supply the parameter lifts used to force that summand's pairing to vanish.

Proof of Theorem 3.1. Suppose the stated data exist and perform the reduction and constructions above. Consider one pure tensor (3.9) contributing to V' . Choose three active positions i, j, k in which

$$B_\nu v_\nu = 0, \quad \delta_\nu v_\nu = 0.$$

On H_ν , the commutator identity is

$$[\delta_\nu, B_\nu] = \delta_\nu B_\nu - B_\nu \delta_\nu = [db_\nu \wedge (-)].$$

It follows that $[db_\nu \wedge (-)]v_\nu = 0$. For a homogeneous D_ν -cycle representative a_ν of v_ν , there is therefore a form w_ν of the same exterior degree with

$$D_\nu w_\nu = -db_\nu \wedge a_\nu.$$

Consequently

$$(D_\nu + t_\nu db_\nu \wedge)(a_\nu + t_\nu w_\nu) = 0 \quad \text{in } \Omega_\nu[t_\nu]/(t_\nu^2). \quad (3.12)$$

Only the vanishing of the indicated classes in H_ν was required; no ordinary d-closed representative is being assumed.

Set all parameters outside this triple equal to zero, and put

$$T_{ijk} = \mathbb{K}[t_i, t_j, t_k] / (t_i^2, t_j^2, t_k^2).$$

Tensor the three lifts (3.12) with the chosen unchanged D_ν -cycle representatives in every other block. The resulting form W_t is a cycle for

$$d(S_0 + t_i b_i + t_j b_j + t_k b_k) \wedge (-).$$

This is an exact cycle equation over T_{ijk} : each perturbation acts in its own block, so the three separate first-order lifts require no additional mixed corrections.

Specialize the actual first identity in (3.7) to these three parameters. It remains an actual boundary identity in forms. In particular, this step does not assert that homology commutes with the possibly nonflat parameter specialization. By (3.8), pairing that identity with W_t gives

$$t_i t_j t_k P(t) = 0 \quad \text{in } T_{ijk}, \quad P(t) = \int U_t \wedge W_t,$$

where U_t is specialized in the same way. The square-free monomials form a $\mathbb{K}$ -basis of T_{ijk} . Multiplication by $t_i t_j t_k$ kills all its positive-degree parameter monomials, so the displayed equation implies

$$P(0) = \int U_0 \wedge (a_1 \otimes \cdots \otimes a_n) = 0.$$

The chosen triple may vary with the pure tensor; the argument applies to each term separately. Summing gives $\int U_0 \wedge V' = 0$. This contradicts (3.11) and the nonzero transverse graph pairing (3.5). $\square$

4 Formal potentials for a single edge

Let Y be a compact connected oriented smooth three-manifold with connected boundary $T \cong S^1 \times S^1$. Fix based coordinate circles α, β on T . We assume that the image of $H^1(Y; \mathbb{C}) \rightarrow H^1(T; \mathbb{C})$ is the line evaluating on α and vanishing on β . These are the boundary data furnished by the edge pieces. All cohomology in this section has complex coefficients. Put $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$, with its invariant nondegenerate pairing $(X, Y) \mapsto \text{tr}(XY)$. Our goal is a formal potential whose actual gradient ideal describes based flat systems with one peripheral holonomy fixed. We also need the first-order change in this potential when that holonomy varies.

Choose closed one-forms ζ, ξ on T with periods

$$\int_\alpha \zeta = \int_\beta \xi = 1, \quad \int_\beta \zeta = \int_\alpha \xi = 0.$$

They may be taken to be bump densities in the two circle coordinates. Their supports are thin strips; choose their common basepoint outside both strips, and choose the coordinate circles through that basepoint. Thus ξ vanishes along α , and ζ vanishes along β . Write

$$\epsilon = \int_T \zeta \wedge \xi \in \{1, -1\},$$

where T has its boundary orientation. Choose a two-form η of integral one supported in a small disk disjoint from the strips. The restriction hypothesis gives a closed extension $\hat{\zeta} \in \Omega^1(Y)$ of ζ : first choose a closed representative with the required boundary cohomology class, and then correct its boundary trace by the differential of an extended function.

4.1 The boundary complex and its pairing

Define a nonunital commutative differential graded algebra C by

$$C^j = \left\{ a \in \Omega^j(Y) : a|_T \in \begin{cases} 0, & j = 0, \\ \mathbb{C}\xi, & j = 1, \\ \mathbb{C}\eta, & j = 2, \\ 0, & j \geq 3. \end{cases} \right\}. \quad (4.1)$$

In degrees at least three the boundary condition is automatic. Multiplication is closed because degree-zero boundary values vanish, $\xi \wedge \xi = 0$, and all other positive-degree products in question vanish on the two-dimensional boundary.

Lemma 4.1. *There are natural isomorphisms*

$$H^1(Y, T) \cong H^1(C), \quad H^2(C) \cong H^2(Y),$$

and $H^0(C) = H^3(C) = 0$. Both middle groups have dimension $n = b_1(Y) - 1 = b_2(Y)$. Every closed one-form in C has zero actual boundary pullback. Integration gives a perfect pairing

$$H^1(C) \otimes H^2(C) \longrightarrow \mathbb{C}.$$

Proof. The quotient of C by the relative de Rham complex has just the two terms $\mathbb{C}\xi$ in degree one and $\mathbb{C}\eta$ in degree two, with zero differential. The connecting map

$$\mathbb{C}\xi \longrightarrow H^2(Y, T)$$

is injective, since $[\xi]$ does not lie in the image of $H^1(Y)$. The connecting map

$$\mathbb{C}\eta \longrightarrow H^3(Y, T)$$

is an isomorphism: under the relative orientation it is boundary integration. The long exact sequence consequently identifies $H^1(C)$ with $H^1(Y, T)$ and $H^2(C)$ with the quotient of $H^2(Y, T)$ by the connecting ξ -line. The ordinary relative sequence identifies this last quotient with $H^2(Y)$. It also gives the asserted vanishing in degrees zero and three. Poincaré–Lefschetz duality gives $\dim H^1(Y, T) = b_2(Y)$, and $\chi(Y) = \frac{1}{2}\chi(T) = 0$ gives $b_2(Y) = b_1(Y) - 1$.

If a closed one-form in C has trace $v\xi$, its boundary cohomology class is both a multiple of $[\xi]$ and an element of the $[\zeta]$ -line. Hence $v = 0$. Under the displayed isomorphisms, the integration pairing is the relative/absolute Poincaré–Lefschetz pairing in degrees one and two, so it is perfect. $\square$

We keep C for the scalar complex. The tensor product $C \otimes \mathfrak{g}$ is a dg Lie algebra, with bracket given by the wedge product combined with the Lie bracket. For homogeneous $a, b \in C \otimes \mathfrak{g}$ whose degrees sum to three, put

$$\langle a, b \rangle = \int_Y \text{tr}(a \wedge b).$$

This pairing satisfies cyclic Stokes. Indeed, when the degrees of a, b sum to two, their boundary product is zero: either one has degree zero or both boundary traces are multiples of ξ . Thus

$$\langle da, b \rangle + (-1)^{|a|} \langle a, db \rangle = 0. \quad (4.2)$$

Invariance of the matrix trace gives the corresponding cyclic identity for the bracket. No assertion that integration of a single degree-two element commutes with the differential is needed; its boundary trace may be a multiple of η .

Choose representatives $i : H^*(C) \rightarrow C$, a projection $\text{pr} : C \rightarrow H^*(C)$, and a degree-minus-one map K such that

$$dK + Kd = \text{id} - i \text{pr}, \quad K^2 = Ki = \text{pr} K = 0, \quad \text{pr} i = \text{id}. \quad (4.3)$$

These maps exist by splitting the vector spaces into boundaries, cohomology representatives, and complements on which the differential is an isomorphism onto boundaries. Extend them by the identity on $\mathfrak{g}$. In particular, the degree-one representatives have zero boundary pullback. Since $H^0(C) = 0$, we have $Kd = \text{id}$ on C^0 .

All series below are formal. Applying one of these linear maps means applying it separately to each coefficient, which is an actual smooth form. No boundedness or analytic convergence of K is required.

4.2 The central slice and its potential

Set

$$H = H^1(C) \otimes \mathfrak{g}, \quad N = \dim H = 3n, \quad S = \mathbb{C}[[h_1, \dots, h_N]],$$

and let h denote the universal element of $H \otimes S$. Its coordinates generate the maximal ideal $\mathfrak{m}$ of S . The recursion

$$s = i(h) - \frac{1}{2}K[s, s] \quad (4.4)$$

has a unique solution in $C^1 \otimes \widehat{\mathfrak{g} \otimes \mathfrak{m}}$: its linear term is $i(h)$, and each higher homogeneous term is determined by previous terms. It satisfies

$$\text{pr} s = h, \quad Ks = 0, \quad KP(s) = 0, \quad P(a) = da + \frac{1}{2}[a, a].$$

Conversely these three slice conditions imply (4.4), by (4.3). Write $s|_T = v(h)\xi$.

We use the Chern–Simons transgression functional [4, Section 3]. For a Lie-algebra-valued one-form a , define

$$\mathcal{C}(a) = \int_Y \text{tr} \left(\frac{1}{2}a \wedge da + \frac{1}{6}a \wedge [a, a] \right), \quad f(h) = \mathcal{C}(s(h)). \quad (4.5)$$

For any coefficient variation $\dot{a}$, integration by parts gives

$$\dot{\mathcal{C}}(a) = \int_Y \text{tr}(\dot{a} \wedge P(a)) - \frac{1}{2} \int_T \text{tr}(a \wedge \dot{a}). \quad (4.6)$$

In the central slice the boundary term is zero.

For a formal form P , its *curvature ideal* means the ideal generated by all its scalar coefficient functions, equivalently by applying arbitrary linear functionals on the vector space of Lie-algebra-valued forms coefficientwise. The next calculation in particular shows that the ideals occurring here are finitely generated.

Lemma 4.2. *The curvature ideal of $s(h)$ is the Jacobian ideal*

$$J = \text{Jac}(f) = \left(\frac{\partial f}{\partial h_1}, \dots, \frac{\partial f}{\partial h_N} \right).$$

Moreover $f \in \mathfrak{m}^3$.

Proof. Put $P = P(s)$ and $r = \text{pr } P \in H^2(C) \otimes \mathfrak{g} \widehat{\otimes} S$. The boundary trace of P is zero, so P lies coefficientwise in $C^2 \otimes \mathfrak{g}$. Bianchi and the projected slice equation give

$$dP = -[s, P], \quad KP = 0, \quad P = i(r) - K[s, P]. \quad (4.7)$$

The operator $K \text{ad}_s$ raises maximal-ideal order. Therefore

$$P = (1 + K \text{ad}_s)^{-1} i(r).$$

Every curvature coefficient lies in the ideal of the finitely many components of r , and the converse holds by applying pr . This is an equality of ideals, without passage to their radicals.

By (4.6), the gradient entries are

$$\frac{\partial f}{\partial h_j} = \left\langle \frac{\partial s}{\partial h_j}, P \right\rangle.$$

Consequently the gradient is a square matrix of formal functions times the vector r . Its constant matrix is the perfect pairing between $H^1(C) \otimes \mathfrak{g}$ and $H^2(C) \otimes \mathfrak{g}$ from Lemma 4.1; it is invertible. This proves the ideal equality. Finally $s = i(h) + O(\mathfrak{m}^2)$ and $di(h) = 0$. Both terms in (4.5) therefore have order at least three. $\square$

4.3 A prescribed first-order boundary holonomy

The central slice fixes the surviving peripheral holonomy to the identity. The vertex construction will vary that holonomy by a square-zero parameter. Its curvature may then have a boundary component, so the next calculation includes that component in the gradient ideal.

Fix $T_0 \in \mathfrak{g}$ and a scalar parameter τ with $\tau^2 = 0$. Put

$$S_\tau = S[\tau]/(\tau^2), \quad u = \tau T_0.$$

The formal topology is the one at $(h, \tau) = 0$. Choose any linear extension

$$K_{\text{ext}} : \Omega^2(Y) \longrightarrow \text{im}(K : C^2 \rightarrow C^1)$$

of the degree-two part of K , and tensor it with $\mathfrak{g}$. Define $s = s(h, \tau)$ by

$$s = i(h) + u\hat{\zeta} - \frac{1}{2}K_{\text{ext}}[s, s]. \quad (4.8)$$

The recursion again determines a unique formal series. It reduces to the central slice when $\tau = 0$, and

$$s - u\hat{\zeta} \in C^1 \otimes \mathfrak{g} \widehat{\otimes} S_\tau, \quad \text{pr}(s - u\hat{\zeta}) = h, \quad K(s - u\hat{\zeta}) = 0.$$

Its boundary value has the form

$$s|_T = u\zeta + v(h, \tau)\xi.$$

The recursion implies $K_{\text{ext}}P(s) = 0$. To see this directly, write its correction as an element ℓ of the original one-form image of K . The contraction identities give $Kd\ell = \ell$; also $di(h) = d\hat{\zeta} = 0$. Applying K_{ext} to the curvature then gives exactly the recursion.

The boundary curvature is generally nonzero off the critical scheme:

$$P(s)|_T = c\zeta \wedge \xi, \quad c = [u, v]. \quad (4.9)$$

In particular it need not satisfy the degree-two boundary condition defining C . The following argument keeps this additional component.

Lemma 4.3. *The function*

$$f_\tau(h) = \mathcal{C}(s(h, \tau)) + \frac{\epsilon}{2} \text{tr}(uv(h, \tau)) \quad (4.10)$$

has internal Jacobian ideal equal to the full curvature ideal of the affine slice. Here the internal derivatives are only the derivatives with respect to the h_j .

Proof. Choose a scalar two-form ρ on Y with boundary trace $\zeta \wedge \xi$, and put

$$P = P(s), \quad P_0 = P - c\rho, \quad r = \text{pr } P_0.$$

Now P_0 has zero boundary trace and lies coefficientwise in $C^2 \otimes \mathfrak{g}$. Applying Stokes to the Lie-algebra-valued Bianchi identity yields

$$\epsilon c = \int_T P = \int_Y dP = - \int_Y [s, P]. \quad (4.11)$$

Thus c is a linear expression in P whose coefficients have positive formal order. Since

$$KP_0 = -cK_{\text{ext}}\rho, \quad dP_0 = -[s, P] - c d\rho,$$

the contraction identity for P_0 gives the exact formula

$$P = i(r) - K[s, P] + c(\rho - dK_{\text{ext}}\rho - Kd\rho). \quad (4.12)$$

The original degree-three part of K is defined on both $[s, P]$ and $d\rho$, since all three-forms satisfy the boundary condition. Substituting (4.11) into (4.12) writes this as

$$P = i(r) + L_s(P),$$

where L_s raises (h, τ) -adic order. Hence

$$P = (1 - L_s)^{-1} i(r).$$

The residual components generate every curvature coefficient. In the other direction, $c = \epsilon^{-1} \int_T P$ and $r = \text{pr}(P - c\rho)$ are linear expressions in curvature coefficients. The residual and curvature ideals are therefore equal, also over the nonreduced ring S_τ . This argument is valid for every allowed extension K_{ext} .

For a variation with u fixed, the boundary term in (4.6) is $-\epsilon \text{tr}(u\dot{v})/2$. The added term in (4.10) cancels it, so

$$\frac{\partial f_\tau}{\partial h_j} = \left\langle \frac{\partial s(h, \tau)}{\partial h_j}, P \right\rangle. \quad (4.13)$$

As in the central case, this is a square matrix times r , with the same perfect constant pairing. The matrix is invertible over S_τ . It follows that the internal Jacobian, residual, and full curvature ideals coincide. In particular the boundary commuting equation $[u, v] = 0$ is included in that ideal; it was not discarded. $\square$

4.4 Based gauge and marked holonomies

The curvature and Jacobian ideals have been identified for both slices. We now identify their quotient schemes with based representations and track the two torus holonomies. Working on Artinian coefficient algebras retains the nilpotent structure used by the ideal calculations.

We use the connection $d + a$ and parallel transport solving $\dot{U} = -a(\dot{\gamma})U$. With the chosen boundary paths this gives

$$G_\alpha = \exp(-u), \quad G_\beta = \exp(-v) \quad (4.14)$$

for a flat boundary connection $u\zeta + v\xi$. Reversing the transport convention changes the corresponding signs throughout.

Let A be a local Artinian complex algebra with residue field $\mathbb{C}$ and maximal ideal $\mathfrak{m}_A$. All connections and gauge transformations in this paragraph reduce to the trivial ones modulo $\mathfrak{m}_A$. The based gauge group consists of transformations equal to the identity at the fixed point of T . On a connection already in the strip boundary form, we subsequently use gauge transformations equal to the identity on all of T .

Lemma 4.4. *The central slice identifies its curvature-zero scheme with the formal scheme of based representations of $\pi_1 Y$ satisfying $G_\alpha = 1$. Over the dual numbers, the affine slice identifies its curvature-zero scheme with the based representations satisfying $G_\alpha = \exp(-\tau T_0)$. These statements hold on all local Artinian coefficient algebras, not only on reduced points.*

Proof. Flat small connections modulo based gauge are equivalent to based small representations by parallel transport. This equivalence can be constructed coefficientwise. A representation defines the flat bundle with its given framing; its small transition functions can be trivialized smoothly by induction over the nilpotent filtration. At each central square-zero stage this is the trivialization of an additive cocycle of smooth functions, which follows from a partition of unity. Conversely, parallel transport is a finite iterated-integral expression at each nilpotent order. Flatness makes it invariant under based homotopies. Two flat connections with the same based monodromy are related by the unique based gauge transformation obtained by comparing their parallel transports.

Suppose first that a based representation has $G_\alpha = \exp(-u)$, where $u = 0$ or $u = \tau T_0$. Put $v = -\log G_\beta$. The two boundary monodromies commute. In the affine case this implies $[u, v] = 0$: conjugating the square-zero exponential gives $(\exp(\text{ad}_v) - 1)u = 0$, and

$$\exp(\text{ad}_v) - 1 = \text{ad}_v \left(1 + \frac{\text{ad}_v}{2!} + \frac{(\text{ad}_v)^2}{3!} + \dots \right),$$

whose factor in parentheses is invertible for nilpotent coefficients. The assertion is automatic when $u = 0$. Hence $u\zeta + v\xi$ is a flat torus connection with exactly the given based monodromy. Based parallel transport gives a gauge transformation normalizing the original boundary connection to this strip form. Its logarithm extends smoothly from the boundary, so the normalization extends over Y .

Now use gauge transformations equal to the identity on T to impose

$$K(a - u\hat{\zeta}) = 0.$$

Their infinitesimal action is $a \mapsto a + d\phi$, with $\phi \in C^0 \otimes \mathfrak{g} \otimes \mathfrak{m}_A$, and $Kd\phi = \phi$. Thus the exact slice coordinate can be killed uniquely, successively in each nilpotent order. For a flat connection in this gauge, setting $h = \text{pr}(a - u\hat{\zeta})$ and applying K or K_{ext} to its zero curvature gives exactly the corresponding recursion. Therefore the normalized connection is $s(h)$ or $s(h, \tau)$.

There is no additional quotient by boundary conjugation. If two normalized boundary connections have the same based monodromy, their coefficients u, v agree by the formal logarithm. A based gauge transformation between those identical boundary connections is the identity everywhere on T , again by parallel transport. The subsequent slice gauge is unique by $Kd = \text{id}$ on C^0 . These constructions are natural under maps of Artinian algebras, proving the assertions about formal schemes, including their nilpotent structure. $\square$

We collect the conclusions, including the first-order coefficient needed later.

Proposition 4.5 (The marked edge potential). *With the boundary polarization fixed above, there is a formal potential $f \in \mathbb{C}[[h_1, \dots, h_N]]$, where $N = 3(b_1(Y) - 1)$, and a slice $s(h)$ such that:*

- (i) *$f = \mathcal{C}(s)$ has order at least three. Its full Jacobian ideal equals the curvature ideal of s . The formal critical scheme $\text{Crit}(f) = \text{Spf}(S/\text{Jac}(f))$ is the based flat-representation scheme with $G_\alpha = 1$, and carries its marked based matrix G_β .*
- (ii) *For each fixed $T_0 \in \mathfrak{g}$ there is an affine slice with boundary $\tau T_0 \zeta + v(h, \tau) \xi$ and adjusted potential*

$$f_\tau = f + \tau b = \mathcal{C}(s(h, \tau)) + \frac{\epsilon}{2} \text{tr}(\tau T_0 v(h, \tau)), \quad \tau^2 = 0.$$

Its internal Jacobian ideal equals its full curvature ideal. Accordingly $\text{Crit}_h(f_\tau)$ is the based flat-representation scheme with $G_\alpha = \exp(-\tau T_0)$. No derivative with respect to τ is imposed in this relative critical scheme.

- (iii) *On the entire, possibly nonreduced, central critical scheme,*

$$b = \epsilon \text{tr}(T_0 v(h, 0)) = -\epsilon \text{tr}(T_0 \log G_\beta) \mod \text{Jac}(f). \quad (4.15)$$

The function b has order at least two. Every flat connection with the indicated normalized boundary value is carried uniquely to the corresponding slice by gauge transformations equal to the identity on T .

Proof. The preceding lemmas prove all assertions except the identification of b and its order. For a general first variation of the two boundary coefficients, the boundary part of (4.6) is

$$-\frac{\epsilon}{2} \text{tr}(u\dot{v} - v\dot{u}).$$

After adding the variation of $\epsilon \text{tr}(uv)/2$, this becomes $\epsilon \text{tr}(v\dot{u})$. Thus the adjusted variation is

$$\dot{f}_\tau = \int_Y \text{tr}(\dot{s} \wedge P(s)) + \epsilon \text{tr}(v\dot{u}). \quad (4.16)$$

Take the coefficient derivative in τ at $\tau = 0$, so that $\dot{u} = T_0$. On $S/\text{Jac}(f)$ the central curvature vanishes as an actual coefficientwise formal form, by Lemma 4.2. The integral term therefore vanishes in that quotient. Equation (4.14) then proves (4.15). This uses equality of ideals and does not pass to the reduced critical locus.

The linear part of the affine slice is $i(h) + \tau T_0 \widehat{\zeta}$, which is closed and has no ξ boundary coefficient. Hence its Chern–Simons expression and its boundary adjustment have joint (h, τ) -order at least three. The coefficient of τ consequently has h -order at least two. All formal exponentials and logarithms used here are defined by their nilpotent truncations on Artinian algebras and their formal limits. $\square$

5 Algebraicity with based holonomy markings

An algebraic power series will mean an element of $\mathbb{C}\langle x_1, \dots, x_N \rangle$, the henselization of $\mathbb{C}[x_1, \dots, x_N]_{(x)}$ inside $\mathbb{C}[[x]]$. We use the edge manifold, boundary coordinates, complex C , and potential f of Proposition 4.5. All ideals in this section are scheme-theoretic ideals; no passage to radicals is understood.

Theorem 5.1 (Algebraic edge potential with markings). *Put $N = \dim(H^1(C) \otimes \mathfrak{sl}_2)$, $R = \mathbb{C}[[h_1, \dots, h_N]]$, and $J = \text{Jac}(f)$. There are an invertible formal coordinate change $x = \varphi(h)$ and an algebraic power series $F(x) \in (x)^3$ such that*

$$f(h) = F(\varphi(h)).$$

Fix a torus basepoint and any finite collection of based loops in Y . For every matrix entry of their holonomies on the central critical scheme $\text{Spf}(R/J)$, there is an algebraic power series in the x -coordinates whose pullback is that entry. In particular this holds for the based β -holonomy. The coordinate change can be chosen to identify these markings on the entire, possibly nonreduced, critical scheme.

The proof has five steps. First, the punctured filling provides an algebraic Darboux chart. Second, the full relative closed-form comparison identifies its symplectic class with the integrated trace on the cochain model. Third, transfer gives the Hamiltonian identity for the given potential f . Fourth, the closed-form homotopy expresses the difference of the two potentials as an element of $\text{Jac}(f)^2$. Fifth, a formal coordinate correction removes that difference while fixing the critical scheme and every marked based holonomy.

5.1 The punctured filling and its algebraic critical chart

Let M be a solid torus with an open ball removed. Glue its torus boundary to ∂Y , with the meridian glued to α , and write

$$Z = Y \cup_{T^2} M, \quad \partial Z = S^2.$$

Choose a point on this sphere and a path to the chosen torus basepoint. The filling kills α in the fundamental group. Thus its based representations are the based representations of $\pi_1 Y$ with $G_\alpha = 1$, and the path identifies their torus markings. The sphere boundary will supply the relative orientation used below. Set $G = SL_2$. For a finite CW complex K , write $\text{Loc}_G(K) = \text{Map}(K_B, BG)$; a superscript fr means that the local system is trivialized at the chosen basepoint. These are derived mapping stacks. In particular, framing is a derived fiber of evaluation at the basepoint, not a quotient by conjugation.

The invariant pairing $\text{tr}(XY)$ is nondegenerate on $\mathfrak{g} = \mathfrak{sl}_2$ and defines a 2-shifted symplectic form on BG . Betti transgression gives a 0-shifted symplectic form on $\text{Loc}_G(S^2)$ [14, Theorem 2.5 and Corollary 2.6(4)]. Restriction

$$\text{Loc}_G(Z) \longrightarrow \text{Loc}_G(S^2)$$

is Lagrangian [3, Definition 2.6, Claim 2.7, and Theorem 2.9]. Here the hypotheses of these constructions have concrete meanings. The manifolds have finite CW type, so the Betti sources are $\mathcal{O}$ -compact and their mapping stacks are derived Artin stacks locally of finite presentation. Integration over the fundamental class is the orientation map. Poincaré–Lefschetz duality supplies the required nondegeneracy for *every* locally constant perfect complex E over every base cdga D , including homotopy-coherent coefficient systems:

$$C^*(Z, S^2; E) \simeq \mathbf{R}\text{Hom}_D(C^*(Z; E^\vee), D)[-3].$$

To see the coefficient scope, local derived duality on contractible stars identifies the oriented interior dualizing complex with $D[3]$ and takes E to $E^\vee \otimes D[3]$. These natural local identifications descend with all higher restriction coherences; relative boundary cells give the displayed equivalence. The

finite constructions in perfect fibers commute with derived base change. This is the compact-manifold application of [3, §§3.1.1–3.1.2], and includes adjoint coefficients. The isotropic homotopy is integration over the relative fundamental chain of (Z, S^2) .

The second Lagrangian is

$$\mathrm{Loc}_G(S^2)^{\mathrm{fr}} \longrightarrow \mathrm{Loc}_G(S^2).$$

Indeed, the based sphere stack is $\{1\} \times_G^{\mathbf{R}} \{1\}$, the derived self-intersection of the identity point in the smooth group G . It is the affine odd vector space with functions $\mathrm{Sym}(\mathfrak{g}^*[1])$, whose generators have degree -1 . On this scheme a form of exterior degree $2 + j$ has internal degree at most $-2 - j$. Thus neither internal degree $-j$, used by a 0-shifted closed two-form, nor internal degree $-j - 1$, used by its nullhomotopy, occurs. The pulled-back form is zero, and its isotropic path is unique up to homotopy. At the unique classical point, the tangent complex of the unframed sphere stack is $C^*(S^2; \mathfrak{g})[1]$; the framed tangent retains exactly the degree-two cohomology summand. Integration and trace pair this summand perfectly with the omitted degree-zero summand. This proves the Lagrangian nondegeneracy condition. There are no other classical points at which to check it.

The shifted intersection theorem therefore equips

$$\mathcal{X} = \mathrm{Loc}_G(Z) \times_{\mathrm{Loc}_G(S^2)}^{\mathbf{R}} \mathrm{Loc}_G(S^2)^{\mathrm{fr}} = \mathrm{Loc}_G(Z)^{\mathrm{fr}} \quad (5.1)$$

with a (-1) -shifted symplectic structure [14, Theorem 2.9]. This is a derived scheme. To see the finite-presentation assertion directly, a compact connected three-manifold with nonempty boundary has a finite spine of dimension at most two: a handle decomposition relative to a collar can be chosen with no absolute three-handles. Collapsing a maximal tree of its one-skeleton gives one vertex, say r edges, and s two-cells. Consequently its framed local-system scheme is the derived fiber

$$G^r \times_{G^s}^{\mathbf{R}} \{1\},$$

where the map records the attaching words. This description also shows that every fixed based holonomy is a regular G -valued function on the classical scheme.

The algebraic Darboux theorem applies over $\mathbb{C}$, at the trivial classical point of $\mathcal{X}$. More precisely, [2, Theorem 5.18(i)] supplies a Zariski local algebraic Darboux chart minimal at that point; in shift -1 , [2, Example 5.15] presents its algebra as

$$A^0[p_1, \dots, p_N], \quad |p_i| = -1, \quad Qp_i = \frac{\partial F}{\partial x_i}, \quad (5.2)$$

where A^0 is smooth and the x_i are étale coordinates. Minimality means that the cotangent differential vanishes at the point [2, Definition 2.13]; in this presentation it is the Hessian of F . Subtracting the constant term therefore gives $F \in (x)^3$. In the completed x -coordinates this is an algebraic power series. Each of the finitely many regular holonomy entries on the classical critical chart lifts through the quotient $A^0 \rightarrow A^0/\mathrm{Jac}(F)$. After shrinking the smooth chart, the lifts are regular, and hence give algebraic power series in the same coordinates. This assertion concerns the matrix entries of holonomy; it makes no assertion that their formal logarithms are algebraic.

The Darboux chart supplies an algebraic potential and regular holonomy functions on its critical scheme. To identify that potential with the particular Chern–Simons function f , we need more than an isomorphism of critical schemes or equality of tangent pairings. The full relative closed-form comparison below will give a difference of potentials in $\mathrm{Jac}(f)^2$. A coordinate correction can then remove that difference while acting trivially on the marked critical scheme. We state the comparison here and give its proof in Appendix A, so that we can first follow its application to the edge complex.

5.2 Comparison of the formal trace forms

The following comparison identifies the full closed form on the cochain model, including the relative integration homotopy. Write $\kappa(u, v) = \text{tr}(uv)$, and denote its 2-shifted closed form on BG by ω_κ . For a nonpositive augmented completed semifree coefficient cdga (R, Q) , let Ω_R^p mean continuous coordinate forms, with their internal grading. These compute the exterior powers of the cotangent complex. If A_X is a commutative cochain model of a finite Betti source, put

$$\text{Cl}_R^2(A_X)^n = \prod_{j \geq 0} (A_X \widehat{\otimes} \Omega_R^{2+j})^{n-j}.$$

Its differential combines $d_X + \mathcal{L}_Q$ and coordinate de Rham differentiation δ_R , with the total-complex signs. The factors of A_X have exterior weight zero. Coordinate differentiation is extended by $\delta_R(c \otimes u) = (-1)^{|c|} c \otimes \delta_R u$. In particular a strict representative has only its $j = 0$ component. Completion in this notation is in the coefficient augmentation ideal; the displayed product is the separate completion in exterior weight.

Proposition 5.2 (Formal trace comparison). *Let X be a finite triangulated space, and let*

$$a_X \in (A_X \otimes \mathfrak{g} \widehat{\otimes} \mathfrak{m}_R)^1, \quad (d_X + Q)a_X + \frac{1}{2}[a_X, a_X] = 0,$$

represent a family of local systems formally near the trivial one. For its evaluation map $\text{ev}_{a_X} : X_B \times \text{Spf } R \rightarrow BG$, the image of $\text{ev}_{a_X}^ \omega_\kappa$ under the Betti coefficient map of [14, §2.1] is the class of*

$$\Theta_X = \frac{1}{2} \kappa(\delta_R a_X, \delta_R a_X), \quad (d_X + \mathcal{L}_Q)\Theta_X = 0, \quad \delta_R \Theta_X = 0, \quad (5.3)$$

in the full complex $\text{Cl}_R^2(A_X)$. This identification is natural for source restrictions, derived coefficient changes, and integration of source cochains.

For an oriented compact three-manifold Z with boundary S , it also identifies the relative-orientation isotropic homotopy with integration of Θ_Z over Z . If $S = S^2$ and the sphere condition is represented by $a_S = \nu b$, where ν is a two-form and b has internal degree -1 , then $\Theta_S = 0$ strictly. Using the zero isotropic path on this sphere model, the resulting intersection form is represented by

$$\frac{1}{2} \int_Z \kappa(\delta_R a_Z, \delta_R a_Z)$$

as a closed two-form of degree -1 .

The proof is given in Appendix A. It constructs the comparison on the formal group nerve, extends it to derived coefficient algebras, and checks source evaluation and relative integration in the full closed-form complex. Throughout the comparison, the marked holonomies are the forward transports fixed in Equation (4.14).

5.3 A model preserving the integrated trace

We identify the cochain model of (5.1) with the edge complex C . On S^2 , choose a closed area form ν . The inclusion $\mathbb{C}\nu$ into the augmented-kernel forms on the based sphere is a quasi-isomorphism. Thus the sphere boundary condition may be imposed by requiring its trace to lie in $\mathbb{C}\nu$, with zero trace in the other degrees. Restriction of unrestricted forms to a boundary is surjective, so this strict fiber computes the required homotopy fiber.

The representatives on the punctured solid torus can be chosen as follows. Extend ξ using a bump density in its longitudinal circle coordinate. Put the puncture and a tube from the support

disk of η to the puncture in the complement of that longitudinal strip. A transverse Thom form of the tube extends η to a closed two-form $\tilde{\eta}$ whose sphere trace is the appropriately oriented ν . Denote the extended one-form by $\tilde{\xi}$. These choices give

$$d\tilde{\xi} = d\tilde{\eta} = 0, \quad \tilde{\xi} \wedge \tilde{\eta} = 0. \quad (5.4)$$

The span of these two forms, with zero multiplication, is a quasi-isomorphic subalgebra of the forms on M with the sphere boundary condition. In fact that complex has one-dimensional cohomology in degrees one and two and no other cohomology, as follows from the relative sequence for (M, S^2) ; the displayed forms represent the two generators.

For gluing, use forms smooth on each piece with matching tangential traces. This piecewise smooth de Rham model computes the cochains of Z ; its integration is the sum over the two pieces, and their torus boundary terms cancel by Stokes. Now glue the small model to the unrestricted forms on Y . Surjectivity of restriction from Y to the torus shows that replacing the model on M by this quasi-isomorphic subalgebra preserves the homotopy pullback. The resulting algebra is exactly C : its torus traces are zero in degree zero, multiples of ξ in degree one, and multiples of η in degree two. Moreover, (5.4) shows that every complementary-degree pairing on the added piece vanishes before integration. The full integrated pairing on the filling is consequently

$$\langle u, v \rangle = \int_Y \text{tr}(u \wedge v) \quad (5.5)$$

on $C \otimes \mathfrak{g}$. The comparison of full closed forms and of the relative integration homotopy in Proposition 5.2 applies to this replacement. It identifies the shifted symplectic structure of $\mathcal{X}$, rather than just its value on tangent cohomology, with this strict integrated form.

Choose the basing path through M outside the longitudinal strip. The degree-one connection on M is a multiple of ξ , so its transport along this path is the identity. The resulting comparison uses the fixed torus framing and preserves based holonomies as matrices, including the chosen β -holonomy.

5.4 Transfer with an arbitrary cochain contraction

The preceding model identifies the integrated trace form and the based holonomies. We next transfer it to the finite coordinate space used by the edge slice, keeping the original contraction. The required output is the Hamiltonian identity for the particular potential f .

Write $H^j = H^j(C) \otimes \mathfrak{g}$, and retain the splitting (i, pr, K) from the edge construction, so

$$dK + Kd = 1 - i \text{pr}, \quad K^2 = Ki = \text{pr} K = 0.$$

No compatibility of K with the pairing is required. Choose bases of H^1, H^2 , and let h and q be their corresponding coordinates of degrees zero and -1 . In the completed semifree algebra $R[q_1, \dots, q_N]$, set

$$s = i_1 h - \frac{1}{2} K[s, s], \quad P = ds + \frac{1}{2} [s, s], \quad r = \text{pr} P, \quad V = (1 + K \text{ad}_s)^{-1} i_2. \quad (5.6)$$

The inverse exists in the (h) -adic topology. The contraction identity and Bianchi identity give $P = i_2 r - K[s, P]$, and hence

$$P = Vr, \quad a = s + Vq, \quad Qh = 0, \quad Qq = -r. \quad (5.7)$$

Here and below the suspension signs in the integrated symplectic form are fixed compatibly with the last convention.

For completeness, these formulas give the entire transferred Maurer–Cartan map. Put $W = dV + [s, V]$. The recursion for V gives $KV = 0$, $\text{pr } V = 1$, and

$$KW = KdV + K[s, V] = V - i_2 + K[s, V] = 0.$$

The columns of W are three-forms in $C \otimes \mathfrak{g}$. They have zero differential by dimension and zero cohomology projection because $H^3(C) = 0$; the contraction identity therefore gives $W = 0$. The bracket of two columns of Vq is a physical four-form and is zero. Equations (5.7) now imply

$$(d + Q)a + \frac{1}{2}[a, a] = 0.$$

In particular there are no omitted terms quadratic in q . The induced map of formal moduli problems has the identity on tangent cohomology, and is a formal equivalence. One can also see this directly by induction over nilpotent coefficient ideals: each square-zero extension has the same cochain deformation and obstruction groups, and the contraction identifies those groups.

Pull back the strict integrated closed form by this map and denote it by ω . It consists of $dh dq$ terms and $q dh dh$ terms. Its constant $dh dq$ matrix is the perfect pairing $H^1 \otimes H^2 \rightarrow \mathbb{C}$, so it is formally nondegenerate. Explicitly, give coordinate forms total parity equal to internal degree plus exterior degree, and extend coordinate differentiation across a physical form c by $\delta(c \otimes u) = (-1)^{|c|} c \otimes \delta u$. The mixed part is then

$$\omega_{\text{mixed}} = - \sum_{i, \mu} \langle \partial_i s, V_\mu \rangle dh_i dq_\mu.$$

The contraction ι_Q is total-even. Since $\iota_Q dq_\mu = -r_\mu$, contraction removes the dq and gives the pairing with $Vr = P$; its value on the $q dh dh$ terms is zero. The variation formula for the central Chern–Simons functional, whose boundary term vanishes, consequently gives the exact identity

$$\iota_Q \omega = \int_Y \text{tr}(\delta s \wedge P) = \delta f. \quad (5.8)$$

The symbol δ differentiates only the formal coordinates. This proves the Hamiltonian identity for the chosen, possibly noncyclic, contraction. Reversing the global symplectic sign reverses both algebraic and Chern–Simons Hamiltonians and has no effect on the statement of Theorem 5.1.

5.5 A closed-form homotopy gives a squared-Jacobian error

We now have two potentials on equivalent minimal formal models: the algebraic Darboux potential and the Chern–Simons potential. The comparison already identifies their critical schemes and based holonomies. We use its additional full closed-form homotopy to place the difference of potentials in the square of the Jacobian ideal. This stronger conclusion permits a coordinate correction that fixes the critical scheme.

Compare the two minimal completed semifree models, the transferred model and the Darboux model (5.2), over the formal completion of $\mathcal{X}$. A map between these models is represented by a formal power-series map: equivalently one uses homotopy transfer and maps of minimal Lie models, or lifts maps of completed semifree algebras order by order. An equivalence has invertible linear part on generators, since the cotangent differentials at the point are zero. It therefore has a formal inverse. Degree considerations make its shape particularly simple:

$$x = x(h), \quad p = B(h)q, \quad (5.9)$$

where the Jacobian of $x(h)$ and the matrix $B(h)$ are invertible. Let $f' = F(x(h))$, and let ω' be the pullback of the canonical Darboux form. Then

$$\iota_Q \omega' = \delta f', \quad (Qq_1, \dots, Qq_N) = \text{Jac}(f') = \text{Jac}(f) = J \subset (h)^2.$$

The first equality is naturality of the Darboux Hamiltonian identity; the equality of ideals follows from the two invertible matrices in (5.9) and from the edge curvature calculation. The comparison is over $\mathcal{X}$, so it also identifies all the marked classical holonomy functions.

Both strict forms represent the same shifted closed-form class. We record explicitly the consequence for the Hamiltonians; equality of the ordinary leading two-form alone would not suffice.

Lemma 5.3 (Completed Cartan calculation). *Suppose that $Qh = 0$, $Qq \in (h)^2$, and that strict closed (-1) -shifted two-forms ω, ω' on $R[q]$ are homotopic as closed forms. If $\iota_Q \omega = \delta f$ and $\iota_Q \omega' = \delta f'$, then, after normalizing constants,*

$$f - f' \in (Qq_1, \dots, Qq_N)^2.$$

Proof. In this proof put $J = (Qq_1, \dots, Qq_N) \subset R$. Use total signs for which $I = \iota_Q$ is even and

$$[I, \delta] = \mathcal{L}_Q, \quad [I, \mathcal{L}_Q] = 0.$$

The closed-form homotopy has components ψ_j , of exterior degree j and internal degree $-j$, for $j \geq 2$. Changing the component signs if necessary, its equations are

$$\omega - \omega' = \mathcal{L}_Q \psi_2, \quad \delta \psi_j = \mathcal{L}_Q \psi_{j+1}.$$

Set

$$S_j = I^j \psi_j, \quad T_j = I^{j-1} \mathcal{L}_Q \psi_j.$$

Commuting δ past the j contractions gives $\delta S_j = T_{j+1} - jT_j$, while $T_2 = \delta(f - f')$. Thus, for every $L \geq 2$,

$$\delta(f - f') = -\delta \sum_{j=2}^L \frac{S_j}{j!} + \frac{T_{L+1}}{L!}. \quad (5.10)$$

Every contraction replaces a dq_i by Qq_i and kills a dh_i . A monomial of ψ_j containing a factors q , b factors dq , and c factors dh satisfies $a + b = j = b + c$, by internal and exterior degree. Hence $a = c$. Survival after j contractions requires $c = 0$, so also $a = 0$ and $b = j$. Consequently S_j is an ordinary function in J^j . For the remainder, the surviving monomials contain either $L + 1$ factors from J , or L such factors and one differentiated factor $\delta(Qq_i)$. Since $Qq_i \in (h)^2$ and $\delta(Qq_i) \in (h)dh$, their coefficient order is at least $2L + 1$. Hence the remainder tends to zero. The sum converges in the complete power-series ring, and (5.10) implies

$$f - f' = -\sum_{j \geq 2} \frac{S_j}{j!} + \text{constant}.$$

The ideal J^2 is adically closed. The normalized constant is zero, proving the claim without an isolated-critical-point hypothesis. $\square$

Lemma 5.4 (Removing the error while fixing the critical scheme). *Let $f \in (h)^3$ and $e \in \text{Jac}(f)^2$. There is a formal automorphism Φ of R , equal to the identity modulo $J = \text{Jac}(f)$, such that*

$$(f + e) \circ \Phi = f.$$

Proof. Write $f_i = \partial_i f$ and $e = \sum_i v_i f_i$, with $v_i = \sum_j a_{ij} f_j \in J$. Since the Hessian of f has entries in (h) , differentiating this expression gives a matrix D with entries in (h) such that

$$\nabla e = D \nabla f.$$

Explicitly, $D_{kj} = \partial_k v_j + \sum_i a_{ij} \partial_k f_i$, which has the stated order. Put $f_t = f + te$. The vector field

$$X_t = -(1 + tD)^{-T} v$$

has entries in $J \subset (h)^2$ and satisfies $df_t(X_t) = -e$. Its formal flow Φ_t , with $\Phi_0 = \text{id}$, exists coefficient by coefficient: each fixed (h) -adic order involves finitely many powers of t , and integration of those polynomials is defined over $\mathbb{C}$. It obeys

$$\frac{d}{dt} \Phi_t^* f_t = \Phi_t^*(e + df_t(X_t)) = 0.$$

Every X_t preserves J , since its coefficients belong to J ; its induced derivation on R/J is zero. Therefore the flow preserves J and is the identity on R/J . Taking $\Phi = \Phi_1$ proves both assertions. $\square$

Proof of Theorem 5.1. Lemma 5.3 gives $f' - f \in J^2$. Apply Lemma 5.4 with $e = f' - f$. Then

$$f = f' \circ \Phi = F \circ x \circ \Phi,$$

so $\varphi = x \circ \Phi$ is the required coordinate change. The original comparison x identifies the critical scheme with the algebraic critical chart and carries its regular holonomy markings to the specified based holonomies. For each such entry H , the additional correction obeys

$$\Phi^* x^* H = x^* H \quad \text{in } R/J,$$

because Φ is the identity on that quotient. The algebraic lifts constructed after (5.2) consequently remain the required lifts after this correction. This proves the marking assertion scheme-theoretically. $\square$

5.6 Polynomial replacement of the active logarithm

Make the fixed coordinate change of the theorem separately on each edge, and again write f for its now algebraic central potential. For an active edge choose an algebraic matrix lift G_β of the based β -holonomy, with $G_\beta(0) = 1$. It suffices to lift its entries; all identities with the actual holonomy below are identities in $R/\text{Jac}(f)$. Set

$$I_\beta = ((G_\beta - 1)_{ij}) \subset R, \quad L_d(G_\beta) = \sum_{k=1}^d \frac{(-1)^{k+1}}{k} (G_\beta - 1)^k.$$

With the connection convention $d + a$ of the edge construction, $G_\beta = \exp(-v)$ on the central critical scheme. Its active coefficient satisfies

$$b \equiv -\epsilon \text{tr}(T_0 \log G_\beta) \pmod{\text{Jac}(f)}.$$

Define the algebraic germ $b_{\text{alg}} = -\epsilon \text{tr}(T_0 L_d(G_\beta))$. For every $d \geq 2$, the (h) -adically convergent logarithm series gives

$$b - b_{\text{alg}} \in \text{Jac}(f) + I_\beta^{d+1} \subset \text{Jac}(f) + I_\beta^3. \quad (5.11)$$

If $b - b_{\text{alg}} = \sum_i c_i \partial_i f + r$, with $r \in I_\beta^{d+1}$, then the square-zero coordinate change $x_i \mapsto x_i - \tau c_i$ transforms $f + \tau b$ into

$$f + \tau(b_{\text{alg}} + r), \quad \tau^2 = 0.$$

Thus the Jacobian part is removed exactly. The remaining error has at least cubic order in the multiplicative holonomy entries; it is this explicit error, rather than an algebraicity assertion about the formal logarithm, that will be used in the vertex matching.

6 Matching the edge potentials at the vertices

We now pass from the cut manifolds to the exact algebraic data of Theorem 3.1. All residual ideals in this section are ideals of formal rings, without passage to their radicals.

Proposition 6.1 (Vertex data). *Suppose that the data of Proposition 2.2 exist, with $m \geq 5$, $p \geq 9$ extra letters, and the finite family of independently conjugated boundary laws described there. Their Brunnian property means that omitting any slot makes the law the identity, as in Equation (2.1). There are a characteristic-zero field K , internal coordinate blocks h_i for the $2m$ edges, and algebraic power-series germs $F_i(h_i)$ and $b_i(h_i)$, the latter for the m active edges, with the following properties. Put*

$$T = K[t_i : i \text{ active}] / (t_i^2 : i \text{ active}), \quad S_0 = \sum_i F_i, \quad S_t = S_0 + \sum_{i \text{ active}} t_i b_i.$$

There are formal parametrizations $H_L(x, t)$ over T and $H_R(y)$ over K whose central tangent maps are injective and have complementary images in the full internal coordinate space. They are smooth graph germs over T and K , respectively, and

$$S_t(H_L) = 0, \quad S_0(H_R) = 0. \quad (6.1)$$

For every three distinct active indices i, j, k ,

$$t_i t_j t_k \in ((\partial_{h_a} S_t)(H_L) : a = 1, \dots, N) \subset T[[x]], \quad (6.2)$$

$$(b_i b_j b_k)(H_R) \in ((\partial_{h_a} S_0)(H_R) : a = 1, \dots, N) \subset K[[y]], \quad (6.3)$$

where N is the total number of internal coordinates. The field can be taken to be $K = \mathbb{C}((\sigma))$. The individual functions can be normalized so that $F_i(0) = b_i(0) = 0$.

We first construct formal sections from the two vertex connection spaces. Their curvature ideals will remain fixed as we normalize the boundary restrictions and correct the sections to make the sums of potentials vanish. We then pass to algebraic edge coordinates and use the boundary laws to obtain the triple memberships. The final step replaces the logarithmic perturbations by algebraic functions and corrects the source section once more.

Orient each Y_i as a boundary face of A . Its orientation as a face of B is the opposite one. Write α_i for its surviving peripheral direction and β_i for the other direction. The plus endpoint is the one at which β_i bounds in the free boundary piece; the minus endpoint is the one at which α_i bounds. An edge is *active* when its plus endpoint is at A . There are m active edges.

Lemma 2.5 gives smooth cuts that miss the locally flat disks. Their complementary pieces are smooth manifolds with corners, so ordinary smooth forms and Stokes' formula apply, with matching pullbacks on their smooth faces at corners.

6.1 The linear restriction maps

Put $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ and use the invariant pairing $\kappa(U, V) = \text{tr}(UV)$. The internal edge space is

$$\mathcal{H} = \bigoplus_i H^1(Y_i, \partial Y_i; \mathbb{C}) \otimes \mathfrak{g}.$$

For a connected oriented three-manifold with connected torus boundary, the natural map from relative degree-one cohomology identifies it with the kernel of restriction to the boundary. Thus these are precisely the internal spaces in Proposition 4.5.

At this linear stage, fixing an evaluation means requiring its tangent variation to vanish. For $P = A, B$, let $\mathcal{V}_P$ be the common kernel in $H^1(P; \mathbb{C}) \otimes \mathfrak{g}$ of evaluation on the extra letters at P and on the α_i whose plus endpoint is P . The degree-one assertion below is the isomorphism $\mathcal{V}_A \oplus \mathcal{V}_B \rightarrow \mathcal{H}$. Later prescribed holonomies need not be the identity; it is their variations that are fixed in this tangent-space statement.

Lemma 6.2. *After fixing the evaluations on the extra letters and on all plus meridians, the direct sum of the two vertex degree-one spaces maps isomorphically, by difference of restrictions, to $\mathcal{H}$. Moreover,*

$$H^2(A; \mathbb{C}) \oplus H^2(B; \mathbb{C}) \longrightarrow \bigoplus_i H^2(Y_i; \mathbb{C}) \quad (6.4)$$

is an isomorphism. In particular each vertex summand maps injectively.

Proof. Lemma 2.6 gives Equation (6.4) and the degree-one restriction isomorphism after the extra evaluations are fixed.

For each edge the image of $H^1(Y_i)$ in the cohomology of its boundary torus is a line. Restrictions from the two vertex sides span $H^1(Y_i)$, and their torus restrictions lie on the two coordinate axes, because the two respective longitudes bound in the boundary pieces. This line must be one coordinate axis. Evaluation on its surviving loop α_i comes only from the plus vertex. Consequently, after the extra evaluations have been fixed, the inverse image of $\bigoplus_i H^1(Y_i, \partial Y_i)$ is exactly the subspace on which all plus evaluations vanish. The asserted restricted map is therefore an isomorphism. Tensoring these statements with $\mathfrak{g}$ proves the result. $\square$

The pairing

$$\bigoplus_i H^1(Y_i, \partial Y_i) \otimes \mathfrak{g} \times \bigoplus_i H^2(Y_i) \otimes \mathfrak{g} \longrightarrow \mathbb{C}, \quad (u, v) \longmapsto \sum_i \int_{Y_i} \kappa(u \wedge v) \quad (6.5)$$

is perfect by Poincaré–Lefschetz duality. We will use both the degree-one isomorphism and this degree-two pairing.

6.2 Marked formal backgrounds and vertex slices

The boundary laws will be tested at a finite family of conjugating matrices whose adjoint operators span every endomorphism of $\mathfrak{g}$. Introduce a formal parameter σ and set $K = \mathbb{C}((\sigma))$. Choose nine matrices in $SL_2(\mathbb{C}[[\sigma]])$, one the identity, all equal to the identity at $\sigma = 0$, whose adjoint operators span $\text{End}_K(\mathfrak{g} \otimes K)$. Equation (6.12) gives an explicit family, and Lemma 6.5 verifies its spanning property. The family is fixed independently of the cuts. Its independent actions on the three slots will detect every triple coefficient in the boundary laws. Assign the eight nonidentity values to eight extra letters at each vertex and set the remaining extra holonomies to the identity. Write $g_s(\sigma) \in SL_2(\mathbb{C}[[\sigma]])$, with $g_s(0) = 1$, for these assigned holonomies. These choices extend to a flat background on the whole ambient bipartite model with trivial meridian holonomies. To see the marking explicitly, choose the paths through the joins as edge paths of the free graph groupoid. The extra loops are its vertex generators, and edge transports can be chosen freely. Choose those transports so that the two torus frames at every join agree with the frames determined by the boundary basing paths. This simultaneously fixes the markings used for holonomy and for the boundary laws.

Initially all constructions take place jointly formally at $\sigma = 0$. The restricted flat backgrounds have coordinates of positive σ order in the central vertex and edge charts. We do not yet invert σ .

Fix $P = A$ or $P = B$, with its basepoint in K_P . In degree zero use smooth $\mathfrak{g}$ -valued functions vanishing at that point, and in positive degrees use unrestricted forms on P . Choose a contraction onto cohomology, denoted by inclusion ι , projection π and homotopy K_P , with

$$dK_P + K_P d = 1 - \iota\pi, \quad K_P^2 = K_P \iota = \pi K_P = 0.$$

Degree-zero cohomology is zero. For $y_{\text{full}} \in H^1(P; \mathbb{C}) \otimes \mathfrak{g}$, the recursion

$$a_P = \iota y_{\text{full}} - \frac{1}{2} K_P [a_P, a_P]$$

defines a unique formal one-form. Write

$$\mathcal{F}_P = da_P + \frac{1}{2} [a_P, a_P], \quad r_P = \pi \mathcal{F}_P.$$

The list r_P has $q_P = \dim(H^2(P; \mathbb{C}) \otimes \mathfrak{g})$ entries, of order at least two. The slice equation and Bianchi identity give

$$\mathcal{F}_P = \iota r_P - K_P [a_P, \mathcal{F}_P] = (1 + K_P \text{ad}_{a_P})^{-1} \iota r_P. \quad (6.6)$$

In particular the curvature ideal is exactly the ideal generated by r_P , and its leading coefficient forms are the selected representatives of $H^2(P) \otimes \mathfrak{g}$. Successively removing exact one-form coordinates by based gauges shows that every based flat formal connection is represented by this slice modulo r_P . These constructions are functorial over nilpotent coefficient rings and pass to their inverse limits. No regular-sequence assertion about r_P is made.

Fix nonzero $T_i \in \mathfrak{g}$ for the active edges. We use the connection convention of Proposition 4.5, in which the strip connection $u\zeta$ has holonomy $\exp(-u)$. Impose on the vertex slice the exact logarithmic holonomies

- $\log g_s(\sigma)$ on the extra letters;
- $-t_i T_i$ on an active plus meridian at A ;
- zero on every plus meridian at B .

There are no parameters on the other plus ends. Logarithmic holonomy along a fixed based loop is a formal function of the off-shell connection, whose linear term is minus integration of the connection along that loop. Lemma 6.2 therefore makes these conditions a smooth formal parameter slice. Keep its other coordinates y_P free. The resulting coefficient rings are

$$R_A = \mathbb{C}[[\sigma, y_A]][t_i : i \text{ active}]/(t_i^2), \quad R_B = \mathbb{C}[[\sigma, y_B]],$$

completed at their displayed joint origins. Write $I_P = (r_P)$ for the substituted residual ideal. It is contained in the square of the joint maximal ideal. Modulo I_P there is an actual based flat system with exactly the prescribed extra and plus holonomies. We keep this list r_P as functions on the parameter domain throughout the subsequent normalizations and section corrections. Thus the flat quotient R_P/I_P stays fixed even when its chosen lifts to forms and edge coordinates change.

For an active edge, the induced holonomies in the flat quotient R_A/I_A satisfy

$$G_{\alpha_i} = \exp(-t_i T_i), \quad G_{\beta_i} = 1.$$

In the flat quotient R_B/I_B they satisfy $G_{\alpha_i} = 1$, while the marked G_{β_i} varies. The central edge potential there contributes $-f_i$ because the boundary orientation is reversed, where f_i is the central potential of Proposition 4.5. These holonomy identities concern the respective flat quotients, not the off-shell lifts.

6.3 Boundary normalization and lifts of the flat quotient

The quotient R_P/I_P carries the flat systems and their group laws. The tensor theorem, however, requires sections in the ambient edge coordinates, where the curvature need not vanish. We first normalize the restrictions on the flat quotient and then lift them to these ambient, or off-shell, coordinates. Changes in the lifts will be measured against the fixed list r_P .

Recall that K_P is a connected sum of p copies of $S^1 \times S^2$ with tubes about an unlink removed. In the unlink exterior choose disjoint proper disks dual to the port meridians, with boundaries the unlink longitudes. In the connected-sum pieces choose disjoint nonseparating spheres dual to the extra S^1 generators. Thin mutually disjoint collars of these disks and spheres carry closed Thom one-forms ρ_ν , normalized to evaluate to one on the corresponding generator. The disk strips meet the port tori in the coordinate densities dual to the port meridians; the sphere strips miss those tori. Choose the basing arcs off the strips. Since the port meridians and extra letters freely generate $\pi_1 K_P$, a flat system with their specified holonomies has the representative

$$a_{K_P} = \sum_{\nu} M_{\nu} \rho_{\nu},$$

where the M_{ν} are the corresponding signed logarithms. Each strip form is closed and their cross wedges vanish, so this connection is flat for arbitrary matrix coefficients. At a plus torus its trace is $u_i \zeta_i$, and at a minus torus its trace is a multiple of ξ_i . The two strip choices are the compatible torus choices of Proposition 4.5. The area representatives can be supported away from both strips. In particular,

$$\mathcal{C}_{K_P}(a_{K_P}) = 0. \quad (6.7)$$

Normalize first over R_P/I_P . Based parallel transport gives a gauge from the restriction of a_P to this strip representative. After this gauge, the flat restriction on every Y_i can be put into the edge slice $s_i(h_i, t_i)$, or $s_i(h_i, 0)$ as appropriate, by a gauge equal to the identity on its torus. This gives coordinates $h_i^P(y_P)$ over R_P/I_P . The normalizations are compatible on intersections of boundary faces: the second gauges are the identity on the tori.

To lift these quotient-ring data to actual formal forms, choose a power-series presentation of the Noetherian complete ring R_P and fix continuous linear division operators for the finite list r_P and for passage to R_P/I_P . Such operators are obtained by formal division, or by choosing compatible filtered linear splittings. Applied coefficient by coefficient, they lift scalar series and write a form-valued discrepancy in I_P as $\sum_a r_{P,a} \gamma_a$ with form-valued series γ_a . Each output coefficient involves only finitely many input coefficients. Thus the same operators apply to the vector spaces of smooth functions and differential forms. This is the actual finite-list ideal membership used below.

Lift the near-identity gauges by first lifting their Lie-algebra-valued logarithms. Smooth collar extension extends each logarithm off the relevant boundary face; the compatibility at the tori allows the extensions to be combined. Lift the coordinates h_i^P as well. Finally change the connection by I_P -multiples of one-forms, extended over a collar of K_P , so that its restriction to K_P is *exactly* the strip representative with the constrained plus coefficients. Denote this trial connection by $\hat{a}_P$. It has the following properties:

1. its curvature belongs to I_P ;
2. its restriction to Y_i equals the specified edge slice modulo I_P ;
3. to first order in the actual residual list, modulo the joint maximal ideal times that list, its curvature representatives are those in Equation (6.6), changed only by exact two-forms.

For the last statement, write an added one-form as $\eta = \sum_a r_{P,a} \gamma_a$ and put $\mathcal{F}(a) = da + \frac{1}{2}[a, a]$ for an intermediate connection a . The source differential treats the functions $r_{P,a}$ of the formal parameters as constants. Hence

$$\begin{aligned} \mathcal{F}(a + \eta) &= \mathcal{F}(a) + \sum_a r_{P,a} d\gamma_a + [a, \eta] + \frac{1}{2}[\eta, \eta] \\ &\equiv \mathcal{F}(a) + \sum_a r_{P,a} d\gamma_a \pmod{\mathfrak{m}_P I_P}. \end{aligned}$$

The congruence is coefficientwise, and $\mathfrak{m}_P$ is the joint maximal ideal: a has positive order and $I_P \subset \mathfrak{m}_P^2$. A gauge with identity value at the joint origin also changes curvature only modulo $\mathfrak{m}_P I_P$. Thus the leading coefficient forms of the fixed residual list change by exact two-forms. This calculation uses a chosen expression for η and requires no independence of the $r_{P,a}$. The lifts have the same first derivatives as the original quotient coordinates because I_P has order at least two.

6.4 The square-ideal and gradient identities

We have lifted the flat boundary restrictions to ambient edge coordinates without changing them modulo the fixed residual ideal. The next calculation measures the error in the sum of their potentials and identifies the restricted gradient ideal. These two statements will permit correction of the sections while preserving their flat-system markings.

Let f_i and b_i^{form} be the edge functions before algebraizing the perturbations. With the orientations already chosen, put

$$S_A = \sum_i f_i + \sum_{i \text{ active}} t_i b_i^{\text{form}}, \quad S_B = - \sum_i f_i.$$

Write H_P for the lifted parametrization consisting of all h_i^P .

Lemma 6.3. *At the joint origin the tangent images of H_A and H_B , with the external parameters fixed, are complementary. For each vertex,*

$$S_P(H_P) \in I_P^2, \quad (\partial_h S_P)(H_P) = M_P r_P, \quad (6.8)$$

where M_P is an actual coefficient matrix whose reduction at the joint origin has full column rank q_P . Consequently the restricted gradient ideal equals I_P .

Proof. The assertion about tangent images is Lemma 6.2: the linearization of restriction to the edge slices is the restriction map into the relative degree-one edge spaces. Thus each parametrization is an immersion and can be written as a graph after a linear projection.

For the first identity, Stokes' formula gives

$$\mathcal{C}_{\partial P}(\widehat{a}_P|_{\partial P}) = \frac{1}{2} \int_P \text{tr}(\mathcal{F}_{\widehat{a}_P} \wedge \mathcal{F}_{\widehat{a}_P}) \in I_P^2.$$

The K_P contribution is zero by Equation (6.7). On an edge interpolate between the trial restriction and its chosen slice. Their difference is in I_P , and every curvature along the interpolation is in I_P . The interior variation of the action is therefore in I_P^2 . On a minus end both boundary traces lie in the ξ_i polarization, so the boundary variation vanishes. The same holds at a frozen plus end, where $u_i = 0$. At an active plus end the trial trace is $u_i \zeta_i$, the slice trace is $u_i \zeta_i + v_i \xi_i$, and the boundary variation is

$$-\frac{\epsilon_i}{2} \text{tr}(u_i v_i).$$

It is canceled exactly by the adjustment in Proposition 4.5. This proves $S_P(H_P) \in I_P^2$ with an actual quadratic expression in the residual list.

The fixed- u variation formula in Equation (4.13) expresses the internal gradient entries as integrations of the slice curvatures against derivatives of the slices. Since those curvatures vanish modulo I_P , all entries belong to I_P . To determine a coefficient matrix, start with Equation (6.6) and the explicit I_P -multiple differences just constructed. At the joint origin, the coefficient of $r_{P,a}$ in the j th gradient entry is, with its boundary-orientation sign,

$$\int_{Y_i} \kappa(\theta_{i,j} \wedge \omega_{P,a}|_{Y_i}). \quad (6.9)$$

Here $\theta_{i,j}$ is the closed relative one-form representing the j th edge tangent and $\omega_{P,a}$ is the chosen closed two-form representative of the residual coordinate. Added leading exact two-forms do not change this number: integrate by parts and use the zero boundary pullback of $\theta_{i,j}$. By Equation (6.4) and the perfect pairing in Equation (6.5), the matrix in Equation (6.9) has full column rank.

Choose such a maximal minor. It is a unit in R_P , so the corresponding gradient entries recover the list r_P by its inverse. This proves equality of ideals. We used the matrix obtained from the curvature expansion, not an inference that the residuals are independent modulo some ideal. Possible syzygies among them have no effect on the argument. $\square$

We record the formal correction argument used at the joint origin.

Lemma 6.4 (Corrections with a fixed residual list). *Let R be a complete Noetherian local $\mathbb{Q}$ -algebra with maximal ideal $\mathfrak{m}_R$, let $I = (r_1, \dots, r_q)$, let $S \in R[[h_1, \dots, h_N]]$, and let $H \in \mathfrak{m}_R^N$. Internal derivatives are taken over R , and substitution at H is $\mathfrak{m}_R$ -adically continuous. Suppose $S(H) \in I^2$ and $(\partial_h S)(H) = Mr$, with a unit full-column minor in M . Assume also that $I \subset \mathfrak{m}_R^2$ and the internal Hessian of S evaluated at H has entries in $\mathfrak{m}_R$. Then there is a convergent correction of H by I -multiples making $S(H) = 0$. It is unchanged modulo I and to first order at the origin; the gradient still generates I . The functions r_a on the parameter domain are held fixed throughout the correction.*

Proof. If the current error is $E = \sum_{a,b} c_{ab} r_a r_b$, use the inverse minor of M to choose a vector δH of I -multiples with $(\partial_h S)(H) \cdot \delta H = -E$. More generally, if the coefficients c_{ab} lie in an ideal Q , choose $\delta H \in QI$. This is a linear calculation using the displayed list and its matrix; it does not require unique expressions for elements of I or I^2 .

An error in $\mathfrak{m}_R^n I^2$ gives a correction in $\mathfrak{m}_R^n I$. The quadratic Taylor term is $\frac{1}{2} \delta H^\top (\text{Hess } S)(H) \delta H$; the factor $1/2$ and all higher Taylor coefficients are defined because R is a $\mathbb{Q}$ -algebra. This term belongs to $\mathfrak{m}_R^{2n+1} I^2$ by the Hessian hypothesis, and higher terms belong to the same ideal because $I \subset \mathfrak{m}_R^2$. At the k th step write $\delta H_k = D_k r$, with every entry of D_k in $\mathfrak{m}_R^{n_k}$, where $n_0 = 0$ and $n_{k+1} = 2n_k + 1$. The Hessian remains in $\mathfrak{m}_R$, so the gradient coefficient matrix changes by a matrix in $\mathfrak{m}_R^{n_k+1}$. Thus both $\sum_k D_k$ and the sequence of gradient coefficient matrices converge. The total correction is $(\sum_k D_k) r \in I$, and the limiting gradient coefficient matrix has the same reduction as the original one, hence retains the chosen unit full-column minor. Continuity now gives $S(H_\infty) = 0$ and $(\partial_h S)(H_\infty) = M_\infty r$. The assumption $I \subset \mathfrak{m}_R^2$ preserves the tangent map. $\square$

Apply Lemma 6.4 to the identities of Lemma 6.3. The central edge functions have order at least three; the other terms carry a parameter t_i . Hence the internal Hessian vanishes at the joint origin. We obtain corrected sections, still denoted by H_A, H_B , with

$$S_A(H_A) = S_B(H_B) = 0, \quad ((\partial_h S_P)(H_P)) = I_P. \quad (6.10)$$

They retain their complementary tangents and their values modulo I_P . In particular they retain exactly the marked edge holonomies on the flat residual schemes.

6.5 Algebraic central coordinates and the order of completion

The corrected sections already satisfy exact vanishing and retain the marked holonomies modulo their residual ideals. We now make the central edge potentials algebraic and move the common background to the origin. The order of this recentering and the subsequent passage to Laurent coefficients is part of the construction.

Apply Theorem 5.1 separately on the edges, initially at the trivial system. Thus, after separate invertible formal changes of the h_i , the central functions are algebraic germs F_i . On their critical schemes the marked β_i holonomies have algebraic lifts G_i in the smooth ambient germs. Choose these lifts with $G_i(0) = 1$. The functions b_i^{form} satisfy

$$b_i^{\text{form}} \equiv -\epsilon_i \operatorname{tr}(T_i \log G_i) \pmod{\operatorname{Jac}(F_i)}. \quad (6.11)$$

Here the sign is the one fixed by $\operatorname{Hol}_{\beta_i} = \exp(-v_i)$. Write the Jacobian-ideal difference as $\sum_a c_{i,a} \partial_{h_{i,a}} F_i$. A separate coordinate change linear in t_i absorbs it: the identity

$$F_i(h_i + t_i c_i(h_i)) = F_i(h_i) + t_i \sum_a c_{i,a}(h_i) \partial_{h_{i,a}} F_i(h_i)$$

is exact because $t_i^2 = 0$. Choosing the sign according to the direction of the change replaces the coefficient by

$$\ell_i(G_i) = -\epsilon_i \operatorname{tr}(T_i \log G_i).$$

The changes are separate in the edge blocks and preserve all equations and ideal memberships on the transformed sections. At parameter zero the changes linear in t_i are the identity.

Let $y_P^0(\sigma)$ and $h_i^0(\sigma)$ be the coordinates of the fixed flat background in these charts. They have positive σ order. Both vertex sections contain the same edge background, because all earlier corrections vanish on their residual schemes. Recenter by $y_P = y_P^0(\sigma) + z_P$ and $h_i = h_i^0(\sigma) + z_i$ *before* passing to Laurent coefficients. Substitution gives continuous maps

$$\mathbb{C}[[\sigma, y_P]] \longrightarrow \mathbb{C}[[\sigma, z_P]], \quad \mathbb{C}[[\sigma, h_i]] \longrightarrow \mathbb{C}[[\sigma, z_i]].$$

For each coefficient in the new internal variables, the defining sum converges σ -adically. We then include the coefficient ring in $K = \mathbb{C}((\sigma))$ and complete in the new internal variables. We rename them y_P, h_i .

This order also describes the changes already used: if ϕ is a separate original formal coordinate change, its recentered expression is

$$z \longmapsto \phi(h_i^0(\sigma) + z) - \phi(h_i^0(\sigma)),$$

formed over $\mathbb{C}[[\sigma, z]]$ first. Later inverses over K may have coefficients with unbounded negative σ orders. They are only composed with centered series, or with series whose constant terms are nilpotent parameter multiples; these substitutions are well defined in $K[[z]][t_i]/(t_i^2)$. In particular, an absorbed Jacobian term gives a separate square-zero change

$$z_i \longmapsto z_i + t_i c_i(z_i), \quad z_i \longmapsto z_i - t_i c_i(z_i) \quad \text{for its inverse.}$$

Its constant displacement is nilpotent. No later substitution shifts an arbitrary Laurent-coefficient series by a nonzero σ -dependent constant, and no operation introduces relations among the external

parameters. Artin approximation is applied only after the final graph identities have been obtained, over the field K as in Lemma 3.2.

The central functions and the G_i are still algebraic germs over K : substitute the background values, which are elements of K , into their finite algebraic presentations. The selected background is critical on every central edge and $G_i(h_i^0) = 1$. Moreover $F_i(h_i^0) = 0$. Indeed ∂F_i vanishes along the formal background curve, so its derivative with respect to σ is zero; its value at $\sigma = 0$ was zero. We retain these normalizations after recentering.

The complementary tangent determinant and the gradient minors have nonzero constant terms at the original joint origin. Their evaluations at the background are units of $\mathbb{C}[[\sigma]]$, and hence are nonzero over K . Thus the two recentered sections are still transverse with complementary dimensions, and their gradient ideals still equal the recentered residual ideals. The latter may now have linear terms; no later argument requires their original quadratic order.

6.6 Conjugated laws on the full residual schemes

The prescribed extra holonomies can be chosen explicitly. In the basis (e, h, f) of $\mathfrak{sl}_2$, put

$$E = \text{ad}(e) = \begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad F = \text{ad}(f) = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}.$$

For $a, b \in \{0, 1, 2\}$ prescribe

$$g_{ab}(\sigma) = \exp(a\sigma e) \exp(b\sigma f). \quad (6.12)$$

The identity conjugator supplies g_{00} and eight extra letters supply the other eight matrices. These choices can be made at both vertices. They require only a finite list of conjugations in the geometric construction, chosen before the cuts are made.

Lemma 6.5. *The nine matrices $\text{Ad}(g_{ab}(\sigma))$ span $\text{End}_K(\mathfrak{g} \otimes K)$.*

Proof. Since $E^3 = F^3 = 0$, the matrices are

$$(1 + a\sigma E + (a\sigma)^2 E^2/2)(1 + b\sigma F + (b\sigma)^2 F^2/2).$$

Order (a, b) lexicographically and vectorize matrices row by row. The determinant of the resulting nine columns is

$$512\sigma^{18}.$$

For a direct calculation, expand in the nine ordered matrices $E^p F^q / (p!q!)$, $0 \leq p, q \leq 2$. Their vectorization determinant is 8. The two evaluation matrices on 0, 1, 2 are Vandermonde matrices of determinant 2, contributing 2^6 to the tensor determinant. The sum of the exponents $p + q$ is 18. The displayed determinant is nonzero in K . $\square$

Lemma 6.6. *For any three distinct active edges, the product of their three source parameters belongs to I_A . At B , every product of one matrix entry from each of their three logarithmic β holonomies belongs to I_B . The same conclusion holds with entries of $G_i - 1$ in place of the logarithmic holonomies.*

Proof. Work in the recentered complete local ring over K of one section, again denoted by R_P , retaining the full square-free parameter algebra at A . Choose off-shell lifts $X_i \in \mathfrak{g} \otimes R_P$ of its marked meridian logarithms on R_P/I_P , and write $\bar{X}_i$ for their quotient images. At A take $X_i = -t_i T_i$; at B take the logarithmic holonomy of the original vertex slice along β_i . Each X_i vanishes at

the background. For three specified edges let J_{ijk} be the ideal in R_P generated by all coordinate products containing one entry from each of X_i, X_j, X_k .

Evaluate the marked boundary words at $\exp(X_i), \exp(X_j), \exp(X_k)$ and the fixed extra holonomies. Modulo I_P these are the actual based holonomies of the flat vertex system, so the boundary laws of Proposition 2.2 make every scalar entry of a law minus the identity lie in I_P . The same entries lie in J_{ijk} : the words are Brunnian in their three core meridians, so omitting any one of the three makes the word the identity. This latter assertion is first an identity for independent formal matrix logarithms and is then substituted into R_P ; it does not require the chosen off-shell connection to be flat.

Their cubic terms, up to the fixed signs and an invertible output conjugation, are the scalar tests of

$$[\text{Ad}(g_a)X_i, [\text{Ad}(g_b)X_j, \text{Ad}(g_c)X_k]], \quad (6.13)$$

with the three conjugators chosen independently. The additional core-word conjugations allowed by the geometric construction equal the identity when the core logarithms vanish, and only change higher terms. In particular, their presence does not assert or require that the globally based word is literally a fixed nested commutator.

By Lemma 6.5, each input of the nonzero trilinear double bracket can independently be acted on by the full endomorphism algebra. To see the spanning assertion directly, choose a nonzero coefficient of this trilinear tensor and use rank-one endomorphisms in its three inputs to select any three desired input coordinates; choose an output functional detecting that coefficient. Every scalar triple coordinate product is therefore a linear combination of the cubic expressions in Equation (6.13).

All higher terms of a Brunnian law lie in $\mathfrak{m}J_{ijk}$, where $\mathfrak{m}$ is the recentered maximal ideal, including the external parameters. If Q is the ideal generated by the scalar law entries, we have

$$Q \subset J_{ijk} \cap I_P, \quad J_{ijk} = Q + \mathfrak{m}J_{ijk}.$$

The module J_{ijk}/Q is finitely generated over the complete Noetherian local ring, so Nakayama's lemma gives $J_{ijk} = Q \subset I_P$. This proves the assertion scheme-theoretically, including all nilpotents and any syzygies in r_P .

At A the quotient logarithms satisfy $\bar{X}_i = -t_i T_i$. Choose a nonzero coordinate of each T_i to deduce $t_i t_j t_k \in I_A$. At B use the corresponding β logarithms. On B 's flat quotient the image of $G_i(H_B)$ is $\bar{G}_i = \exp(\bar{X}_i)$. Every entry of $\bar{G}_i - 1$ is therefore in the ideal generated by the entries of $\bar{X}_i$. The triple products of the lifted entries $G_i - 1$ consequently belong to I_B as well. All earlier section corrections were the identity modulo the residual ideals, so the same conclusions hold on the corrected sections. $\square$

6.7 Polynomial perturbations and the final graph families

The triple-ideal identities are now established on the full residual schemes. The remaining task is to replace the logarithmic perturbation by an algebraic function without losing exact vanishing or gradient generation. Algebraic lifts of the holonomy matrices suffice: a quadratic polynomial in those matrices has a controlled cubic error, which can be removed on the source graph by a nilpotent correction.

For an active edge replace the logarithmic function by the polynomial

$$b_i(h_i) = -\epsilon_i \text{tr} \left(T_i \left((G_i(h_i) - 1) - \frac{1}{2}(G_i(h_i) - 1)^2 \right) \right). \quad (6.14)$$

These are algebraic germs over K , with zero constant term at the recentered background. We check the exact effects of this replacement.

On the plus section, setting $t_i = 0$ gives, modulo I_A , a central critical system on edge i with both peripheral holonomies trivial: α_i is constrained to be trivial and β_i bounds in K_A . The marked algebraic holonomy therefore satisfies

$$G_i(H_A) - 1 \in (I_A, t_i) \quad (6.15)$$

entry by entry. This is equality in the quotient ring, not just evaluation at its closed points. The coordinate change used to absorb the difference in Equation (6.11) is immaterial after setting $t_i = 0$.

Let $R_i = \ell_i(G_i) - b_i$. It has order at least three in the entries of $G_i - 1$, and each internal derivative has order at least two in those entries. Since $t_i^2 = 0$, Equation (6.15) gives

$$t_i R_i(H_A) \in t_i(I_A, t_i)^3 = t_i I_A^3, \quad (6.16)$$

$$t_i(\partial_h R_i)(H_A) \in t_i(I_A, t_i)^2 = t_i I_A^2. \quad (6.17)$$

Thus the new total function on the existing plus section lies in $\mathfrak{t}I_A^2$, where $\mathfrak{t} = (t_i : i \text{ active})$, and its restricted gradient still has the form $(M_A + \mathfrak{t}Q)r_A$ with a unit full-column minor.

The correction at the joint origin used the vanishing internal Hessian and gained powers of the joint maximal ideal. After recentering, the Hessian may be nonzero. Here the error instead contains a factor from the nilpotent parameter ideal, and the following correction gains powers of that ideal while keeping the same residual list.

Lemma 6.7 (Correction over the full parameter algebra). *Let I be a finite set of cardinality m , put $T = K[t_i : i \in I]/(t_i^2 : i \in I)$, let $J = (t_i : i \in I)$, and work in $T[[x]]$. Fix a finite residual list r , with ideal $\mathcal{I} = (r)$. Suppose a formal section H and a formal potential S satisfy*

$$S(H) \in J\mathcal{I}^2, \quad (\partial_h S)(H) = Mr,$$

where M has a unit full-column minor. Then a displacement of H in $J\mathcal{I}$ makes $S(H) = 0$ identically and keeps the restricted gradient ideal equal to $\mathcal{I}$. If the original central parametrization has an injective tangent map, the corrected section is still a smooth formal graph over T .

Proof. Keep the functions r fixed on the parameter domain. Suppose at one step the error is in $J^q \mathcal{I}^2$, $q \geq 1$. Write it as $\sum_{a,b} c_{ab} r_a r_b$, with $c_{ab} \in J^q$. Using the inverse full-column minor of the restricted gradient matrix, choose a displacement $\Delta H \in J^q \mathcal{I}$ whose linear Taylor term cancels this error. The Taylor remainder belongs to

$$(\Delta H)^2 \subset J^{2q} \mathcal{I}^2 \subset J^{q+1} \mathcal{I}^2.$$

This uses no vanishing of the internal Hessian. The gradient changes by an element of $J^q \mathcal{I}$, so its new coefficient matrix is $M + J^q Q$ for some matrix Q . Its chosen minor remains a unit because J is nilpotent. Iterating terminates, since $J^{m+1} = 0$, and preserves gradient generation throughout. Every correction vanishes at $t = 0$. A fixed linear projection that was invertible on the central tangent therefore still has a unit relative Jacobian. Formal inversion over T , fixing all parameters, gives the asserted graph. The inverse reparametrization also pulls back every residual membership witness. $\square$

Apply Lemma 6.7 over $T[[y_A]]$, with $J = \mathfrak{t}$, $\mathcal{I} = I_A$, and the fixed residual list r_A . The preceding estimates give an error in $\mathfrak{t}I_A^3 \subset \mathfrak{t}I_A^2$ and a unit full-column gradient minor against that same list. The section is centered modulo $\mathfrak{t}$: $H_A(0, 0) = 0$, so $H_A \in (y_A, \mathfrak{t})^N$. Formal substitution is continuous even with its nilpotent constants: write $H_A = H_A(y_A, 0) + \eta$, where $\eta \in \mathfrak{t}T[[y_A]]^N$; expansion in η is finite and substitution of $H_A(y_A, 0)$ is centered. The central tangent remains injective by

Lemma 6.2 and the preceding coordinate changes. Thus the lemma gives a corrected source section H_L with exact vanishing for the polynomial perturbations. The corrections lie in $\mathfrak{t}I_A$ and preserve the gradient ideal I_A . The ideal $\mathfrak{t}$ is nilpotent of exponent at most $m + 1$; its square is not set to zero. In particular, mixed parameter monomials are retained at every step. A nonzero Hessian of a recentered central potential causes no difficulty because the Taylor remainders gain powers of $\mathfrak{t}$.

The minus function is still S_0 up to its harmless overall sign, so the minus section needs no additional correction. Denote it by H_R . Every weight in Equation (6.14) has zero constant term in the entries of $G_i - 1$. Lemma 6.6 therefore gives $(b_i b_j b_k)(H_R) \in I_B$ for distinct active indices. The source parameter products remain in I_A . Since the final restricted gradients generate these same ideals, these are exactly Equations (6.2) and (6.3).

For completeness, the corrected source is a smooth graph over the *full* ring T , rather than just on its reduced central fiber. Choose a linear projection ρ on the internal coordinate space which is invertible on the central tangent of the source. The Jacobian of $y_A \mapsto \rho H_L(y_A, t)$ is invertible modulo (t) and hence is a unit over $T[[y_A]]$. The constant term of ρH_L can lie in the nilpotent ideal (t) ; translation by that constant is continuous for the (y_A, t) -adic topology, which here agrees with the y_A -adic topology. The formal inverse theorem over T reparametrizes this map as

$$H_L(x, t) = (x, \Psi(x, t)),$$

while leaving the parameters fixed. All memberships pull back under this same T -algebra automorphism. At $t = 0$ the last corrections vanish, so complementarity with H_R is preserved. No flatness of $T[[y_A]]/I_A$ over T is required.

Proof of Proposition 6.1. Use the smooth cut data and choose the finite extra-letter background in Equation (6.12). Lemma 6.2, the based connection slices, and Lemma 6.3 give complementary formal sections, square-ideal errors, and gradient generation by the actual residual lists. Lemma 6.4 gives exact vanishing. Apply the separate marked algebraic edge-coordinate comparisons, absorb the Jacobian terms, and recenter before extending to $K = \mathbb{C}((\sigma))$. Lemma 6.6 supplies the exact source and target triple-product memberships. Finally Equations (6.14)–(6.17) and the nilpotent correction produce algebraic F_i, b_i and the required final graphs. The preceding graph reparametrization supplies precisely the stated smooth families. This proves all assertions. $\square$

Corollary 6.8. *The cut data of Proposition 2.2, with $m \geq 5$ and the indicated finite law list, supply all the hypotheses of Theorem 3.1.*

Proof. Take the field, separate algebraic edge functions and graph families of Proposition 6.1. Equations (6.1)–(6.3) are the exact equations and memberships required there. The graphs may mix the edge coordinates, while the fixed potentials remain separated by edge. Their external parameter ring is the complete square-free algebra T , and their central tangent spaces are complementary. Ordinary algebraic approximation and subsequent spreading out are applied only at this final stage, as in the proof of Theorem 3.1; they are not applied to the earlier arbitrary formal potentials or to a Laurent series term by term. $\square$

7 The obstruction

Proof of Theorem 1.1. Choose $m \geq 5$ and the finite extra-letter and law data used in Proposition 6.1. Before invoking Assumption 2.1, the geometric construction produces the fixed immersed disk problem of Lemma 2.4 in

$$E = X \setminus \bigcup_j \text{int } \nu(S_j).$$

Suppose that this particular family admits simultaneous pairwise disjoint locally flat replacements with the same boundary maps and induced boundary framings. The proof of Proposition 2.2 then constructs the required smooth vertex and edge pieces. Proposition 4.5 and Theorem 5.1 give their separate algebraic central potentials and marked holonomy germs.

Proposition 6.1 then produces complementary formal graphs over the full parameter algebra

$$T = K[t_1, \dots, t_m]/(t_1^2, \dots, t_m^2),$$

whose parameters are individually square-zero, with the source and target triple-product gradient-ideal memberships of Theorem 3.1. There are at least five active factors. That theorem gives a contradiction.

Consequently this fixed disk problem has no such embedded replacement. Its ambient manifold E is compact, connected, oriented, and smooth. Lemma 2.4 verifies all the displayed equivariant intersection, self-intersection, and framing conditions for its immersed disk and sphere inputs. Taking $M = E$ gives the asserted example. $\square$

Remark 7.1 (Scope of the geometric conclusion). The examples lie in the compact orientable smooth class. They therefore refute the corresponding universal assertion for all topological four-manifolds, including versions allowing noncompact and nonorientable manifolds. The markings used to construct the example impose no additional condition on the hypothetical replacements: only their boundary maps and induced boundary normal framings have been prescribed, with no relative homotopy requirement.

The contradiction already uses the disk conclusion of the unrestricted assertion. Allowing self-intersections and mutual intersections among the hypothetical output dual spheres does not remove it. We make no separate counterexample claim here for the ordinary surgery sequence, the simple surgery sequence on finite simple Poincaré pairs, or five-dimensional topological s -cobordism. Such a conclusion would require an additional implication from the independently formulated assertion to the disk replacement problem treated here. In particular, finiteness of a Poincaré complex is not being substituted for simple Poincaré duality.

A The full relative trace comparison

We prove Proposition 5.2, including its compatibility with the relative integration homotopy. This is the comparison used in Section 5 to identify the shifted symplectic structure on the punctured filling with the integrated trace on the edge complex.

We retain the closed-form complex and total signs of Section 5.2. There are three differentials: the source differential d_X , the coefficient cdga differential Q , and coordinate de Rham differentiation δ_R . On coordinate forms Q acts by $\mathcal{L}_Q$. Source and simplex forms contribute to internal degree but have exterior weight zero; the coordinate forms supply exterior weight. Coefficient completion and the product over exterior weights are both retained throughout the argument.

Proof of Proposition 5.2. 1. The target cocycle. We first construct the target form on a presentation of the formal classifying stack. Let $\widehat{G}$ be the formal group of G at the identity. Here the pointed completion of BG means the deformation functor with a specified identification of its reduction with the trivial object; its automorphisms reduce to the identity. This pointed completion is $B\widehat{G}$, presented by the group nerve $\widehat{G}^\bullet$. Its term $\widehat{G}^n$ is formally smooth, with completed coordinate algebra

$$R_n = \mathbb{C}[[x_1, \dots, x_{n \dim G}]].$$

Continuous coordinate forms on these algebras therefore compute derived forms on the nerve.

We use backward transition elements for the group nerve. More explicitly, if U_{ij} is the forward transport from vertex i to vertex j for the connection $d + a$, then $U_{ik} = U_{jk}U_{ij}$. The nerve labels $h_{ij} = U_{ij}^{-1}$ therefore satisfy $h_{ik} = h_{ij}h_{jk}$. For a left Maurer–Cartan form $a = \gamma^{-1}d\gamma$, these labels are $h_{ij} = \gamma(i)^{-1}\gamma(j)$, whereas the forward transport is $U_{ij} = \gamma(j)^{-1}\gamma(i)$. Thus Getzler’s interval labeled by $\exp(x)$ has form $x dt$, and its forward holonomy is $\exp(-x)$, as in the edge construction. This is the opposite-arrow convention for describing the same local system; inversion is not being regarded as a homomorphism from a noncommutative group to itself. All based holonomy markings below remain the forward transport matrices.

For an ordinary nilpotent ideal $\mathfrak{m}$, the Lie algebra $\mathfrak{g} \otimes \mathfrak{m}$ is nilpotent. Getzler’s simplicial Maurer–Cartan space is

$$\mathrm{MC}_\bullet(\mathfrak{g} \otimes \mathfrak{m}) = \mathrm{MC}(\mathfrak{g} \otimes \mathfrak{m} \otimes \Omega_{\mathrm{poly}}^*(\Delta^\bullet)).$$

Its Dupont–gauge subspace $\gamma_\bullet$, defined using the simplicial de Rham contraction of [5], is the nerve of the Campbell–Hausdorff group in this degree-zero case, and its inclusion in $\mathrm{MC}_\bullet$ is a natural homotopy equivalence [11, §1, Theorem 5.4, and Corollary 5.11]. Applying this construction to the universal n -tuple, first at each finite augmentation order and then taking the inverse limit, gives compatible flat simplices

$$a_n \in (\mathfrak{g} \otimes \mathfrak{m}_{R_n} \hat{\otimes} \Omega_{\mathrm{poly}}^*(\Delta^n))^1.$$

The compatibility here is that of Getzler’s simplicial construction, including every face and degeneracy.

Differentiate only in the R_n directions and set

$$\beta_n = \tfrac{1}{2}\kappa(\delta_{R_n} a_n, \delta_{R_n} a_n). \quad (\mathrm{A.1})$$

Polynomial simplex forms have internal degree but exterior weight zero in this expression. Differentiating the Maurer–Cartan equation gives the covariant-closedness equation for $\delta_{R_n} a_n$. Invariance of κ cancels the two bracket terms and yields

$$d_\Delta \beta_n = 0, \quad \delta_{R_n} \beta_n = 0.$$

These identities and simplicial compatibility make $\beta_\bullet$ a Thom–Whitney descent cocycle of internal degree two and exterior weight two on the smooth formal nerve. All its higher closed-form components are zero. Equivalently, integration over the simplices gives a normalized descent cocycle; its descent differential is zero by Stokes’ formula.

2. The canonical closed-form class. We identify its class with the restriction of ω_κ . The cotangent complex of $B\hat{G}$ is the coadjoint bundle $\mathfrak{g}^*[-1]$. Consequently

$$R\Gamma(B\hat{G}, \bigwedge^p \mathbb{L}_{B\hat{G}}) \in D^{\geq p}.$$

Indeed, $\bigwedge^p \mathbb{L}$ is the bundle $\mathrm{Sym}^p(\mathfrak{g}^*)[-p]$, and derived invariants of a bundle in degree zero have no negative cohomology. Applied to the exterior weight filtration of the closed two-form complex, this connectivity gives

$$H^2 \mathrm{Cl}^2(B\hat{G}) \simeq \mathrm{Sym}^2(\mathfrak{g}^*)^{\hat{G}}, \quad H^1 \mathrm{Cl}^2(B\hat{G}) = 0. \quad (\mathrm{A.2})$$

For the j -th higher component, the relevant internal degree would be $2 - j$, whereas weight $2 + j$ starts in degree $2 + j$. This is a statement about the derived form complexes and their classes, independent of the terms in a chosen descent resolution.

Linearizing the nerve at the identity gives the Dold–Kan nerve of $\mathfrak{g}[1]$. Formula (A.1) linearizes to its quadratic two-form with associated symmetric bilinear form κ . Here we use the usual normalized

simplex integration and shuffle convention: the two shuffles in the polarization cancel the factor $1/2$. This is precisely the underlying bilinear form of ω_κ [14, §1.2, classifying stacks]. Equation (A.2) identifies their full closed-form classes, and makes the comparison homotopy unique up to homotopy.

3. Derived coefficients. We now give the comparison on derived coefficients, including its map on linear fibers. By an Artin test algebra here we mean a nonpositive local augmented cdga B over $\mathbb{C}$, with nilpotent augmentation ideal $\mathfrak{m}_B$. Set

$$L_B = \mathfrak{g} \otimes \mathfrak{m}_B, \quad \Omega_q = \Omega_{\text{poly}}^*(\Delta^q), \quad H_q(B) = \exp Z^0(L_B \otimes \Omega_q).$$

The exponential uses the finite Campbell–Hausdorff series. These groups form a simplicial group $H_\bullet(B)$. They compute the derived evaluation of $\widehat{G}$ on B : in exponential coordinates a map from its completed smooth coordinate algebra sends each coordinate to a degree-zero cycle in $\mathfrak{m}_B \otimes \Omega_q$. Equivalently one uses the nonpositive truncation of this polynomial simplicial frame. Consequently the diagonal of the bisimplicial nerve $N_n H_q(B)$ computes $B\widehat{G}(B)$.

Apply the ordinary degree-zero Getzler construction to an n -tuple in $H_q(B)$. The inclusion of the degree-zero Lie algebra $Z^0(L_B \otimes \Omega_q)$ into $L_B \otimes \Omega_q$ gives a flat element

$$A_{n,q} \in \text{MC}(L_B \otimes \Omega_q \otimes \Omega_n).$$

The coefficient cycles have zero total differential, so this is the Maurer–Cartan equation for the full displayed tensor product. Getzler’s compatibility in n and functoriality in the coefficient Lie algebra give compatibility in both simplicial directions, including all faces and degeneracies. Restriction to the diagonal simplex gives a natural simplicial map

$$\Phi_B : \text{diag } N_\bullet H_\bullet(B) \longrightarrow \text{MC}_\bullet(L_B), \quad (\Phi_B)_q = \Delta_q^* A_{q,q}. \quad (\text{A.3})$$

Here is a direct verification that this particular map is an equivalence. Consider a central step $B \twoheadrightarrow B'$ in the filtration by powers of $\mathfrak{m}_B$, with kernel I satisfying $\mathfrak{m}_B I = 0$, and write $V = \mathfrak{g} \otimes I$. The map $Z^0(L_B \otimes \Omega_q) \rightarrow Z^0(L_{B'} \otimes \Omega_q)$ is surjective. Indeed, lift a closed element to w . Its differential Dw is a closed degree-one element of $V \otimes \Omega_q$; the polynomial Poincaré lemma and nonpositivity give $H^1(V \otimes \Omega_q) = H^1(V) = 0$. Subtracting a degree-zero primitive of Dw gives a closed lift. Thus the induced map of simplicial groups is degreewise surjective, with additive kernel

$$K_q = Z^0(V \otimes \Omega_q).$$

Its classifying-space homotopy fiber is $BK_\bullet$. On the Maurer–Cartan side the surjection is a Kan fibration with fiber $\text{MC}_\bullet(V)$, by [11, Proposition 4.7]. We use homotopy fibers of the classifying spaces here; no Kan-fibration assertion about an arbitrary diagonal map is needed.

The restriction of Φ_B to these fibers can be checked explicitly. For the abelian complex V , put

$$C_q = (V \otimes \Omega_q)^0, \quad M_q = Z^1(V \otimes \Omega_q) = \text{MC}_q(V).$$

There is a degreewise exact sequence of simplicial vector spaces

$$0 \longrightarrow K_\bullet \longrightarrow C_\bullet \xrightarrow{D} M_\bullet \longrightarrow 0.$$

Let $(EK)_q = K_q^{q+1}$ be the homogeneous bar construction, with diagonal translation action of K_q . Its quotient is the diagonal inhomogeneous nerve BK . The map

$$\Psi_q : (EK)_q \longrightarrow C_q, \quad (k_0, \dots, k_q) \longmapsto \sum_{j=0}^q t_j k_j$$

is simplicial and restricts to the identity on the diagonal copy of K_q . Since $Dk_j = 0$, its quotient map is $D\Psi_q = \sum_j dt_j k_j$. In the inhomogeneous coordinates $x_i = k_i - k_{i-1}$, this is exactly the abelian Getzler simplex after the diagonal restriction in (A.3). Thus we have a commuting diagram of exact simplicial vector spaces

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K & \longrightarrow & EK & \longrightarrow & BK & \longrightarrow & 0 \\ & & \parallel & & \downarrow \Psi & & \downarrow \Phi_V & & \\ 0 & \longrightarrow & K & \longrightarrow & C & \xrightarrow{D} & M & \longrightarrow & 0. \end{array}$$

Both middle terms are contractible. This can be seen without choosing a contraction of V : for a simplicial interval parameter $\sigma : [q] \rightarrow [1]$, multiplication by $\sum_{\sigma(j)=1} t_j$ gives a simplicial homotopy from zero to the identity on C_q . On $(EK)_q$, use $(k_j)_j \mapsto (\mathbf{1}_{\sigma(j)=1} k_j)_j$. The connecting maps of the two exact sequences therefore identify Φ_V with the identity suspension of K . In particular it is a weak equivalence, and under the compatible integration and suspension identifications it is the canonical map from the bar model of V to the model of $V[1]$. This statement uses those connecting-map conventions, not an unsigned equality of raw simplex integrals. Induction over the finite central filtration, starting at $B = \mathbb{C}$, now proves that Φ_B is an equivalence. The construction is natural in B ; the usual nilpotent Maurer–Cartan invariance then also gives its compatibility with quasi-isomorphisms. This is a proof of the dg extension, rather than an assertion that the ordinary coefficient-ring statement of [11, Proposition 4.9] already states that extension.

4. Compatibility of the closed cocycle. We finally check the closed cocycle under this very comparison. Do this before simplex integration, in the Thom–Whitney mixed complexes, and use a completed cofibrant coefficient resolution when computing derived forms. A tuple in $H_q(B)$ gives the substitution of the universal nerve coordinates by closed elements of $\mathfrak{m}_B \otimes \Omega_q$. Pull back $\beta_n = \frac{1}{2}\kappa(\delta a_n, \delta a_n)$ by that substitution, and apply the coefficient morphism

$$\mathrm{DR}(B \otimes \Omega_q) \longrightarrow \mathrm{DR}(B) \otimes \Omega_q.$$

As throughout this proof, this notation means derived de Rham objects computed on the chosen resolutions. The morphism puts the polynomial simplex forms in exterior weight zero. In particular coordinate differentiation still differentiates every B -dependent coefficient of the logarithms of the tuple; those coefficients are not held constant. Naturality of coordinate differentiation gives precisely

$$\frac{1}{2}\kappa(\delta_B A_{n,q}, \delta_B A_{n,q}).$$

Both simplex directions have exterior weight zero. Diagonal restriction commutes with their internal differential and with δ_B , with the total signs already specified. Consequently

$$\Delta_q^* \left(\frac{1}{2}\kappa(\delta_B A_{q,q}, \delta_B A_{q,q}) \right) = \frac{1}{2}\kappa(\delta_B(\Phi_B)_q, \delta_B(\Phi_B)_q)$$

as full mixed cocycles. The higher closed-form components of these representatives are all zero; no exterior weights have been dropped. Faces, degeneracies, coefficient maps, and their homotopies commute with this equality. The standard Thom–Whitney-to-normalized comparison therefore gives equality of the full closed-form classes. We do not identify integration over a product of simplices with integration over its diagonal. This establishes the derived pullback of the class already constructed on the smooth nerve, together with its natural comparison homotopy. Passing continuously through the nilpotent coefficient quotients gives the same assertion for a completed semifree R , where continuous coordinate forms compute the derived forms.

5. Source evaluation and relative integration. We next carry out evaluation on the source. Choose a finite triangulation K of X . A compatible family of flat simplices

$$a_\sigma \in \text{MC}(\mathfrak{g} \otimes \mathfrak{m}_R \hat{\otimes} \Omega_{\text{poly}}^*(\Delta^{\dim \sigma})), \quad \sigma \in K,$$

is a simplicial map from K to the Maurer–Cartan model of $B\hat{G}$. Including simplicial test parameters gives its families and all their homotopies. Evaluation is the evaluation of these simplicial maps. Pulling back the target cocycle therefore assigns to each σ the expression $\frac{1}{2}\kappa(\delta_R a_\sigma, \delta_R a_\sigma)$.

Before totalizing this diagram, apply the coefficient map

$$\kappa_{\text{Spf } R, X_B} : \text{DR}(\text{Spf } R \times X_B) \longrightarrow \text{DR}(\text{Spf } R) \hat{\otimes} R\Gamma(X_B, \mathcal{O}_{X_B})$$

used in [14, §2.1, before Definition 2.3]. It keeps source functions and cochains in exterior weight zero. Thus the polynomial simplex forms above belong to the source cochain factor, while δ_R supplies every exterior weight in the coefficient form. Totalizing afterwards produces exactly Θ_X in (5.3). We have proved the equality of full closed-form classes

$$\kappa_{\text{Spf } R, X_B} \text{ev}_{a_X}^*[\omega_\kappa] = [\Theta_X].$$

All these operations are performed weight by weight and are continuous in the coefficient ideal. They then define the displayed completed closed-form complex.

For a smooth triangulated manifold, the same statement uses smooth forms through the multiplicative comparison

$$\Omega_{\text{sm}}^*(X) \longrightarrow \Omega_{\text{ps}}^*(K) \longleftarrow \Omega_{\text{poly}}^*(K),$$

where the middle term consists of compatible piecewise smooth forms. Both arrows are quasi-isomorphisms and commute with restriction. Integration of forms over the oriented simplices is the same on all three models, including their boundaries. The construction therefore commutes with the orientation cochain maps, as well as with any multiplicative replacement carrying those maps.

Finally let $\partial Z = S$. Write I_Z and I_S for integration over compatible fundamental chains. With $\partial[Z] = [S]$ and the suspension convention for transgression, the equations in (5.3) give

$$h_Z = I_Z \Theta_Z, \quad \mathcal{L}_Q h_Z = I_S \Theta_S, \quad \delta_R h_Z = 0.$$

Thus h_Z , of internal degree -1 , is the relative closed-form homotopy from zero to the pulled-back boundary form. This is the homotopy obtained by applying the orientation construction to evaluation [3, Definition 2.6 and Claim 2.7]. The comparison homotopy on $B\hat{G}$ is carried through the same evaluation and relative-chain maps. It consequently compares these relative homotopies together with their endpoints.

On the sphere condition $a_S = \nu b$, every term of $\kappa(\delta_R a_S, \delta_R a_S)$ contains $\nu \wedge \nu = 0$. Its full closed-form representative and its chosen path are therefore zero. Subtracting this zero path in the shifted Lagrangian intersection construction leaves h_Z , now closed of degree -1 , as asserted. $\square$

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